BT1010 Introduction to Life Sciences



Lecture 5: Techniques used in cell and molecular biology and their diagnostic applications

06/11/2025

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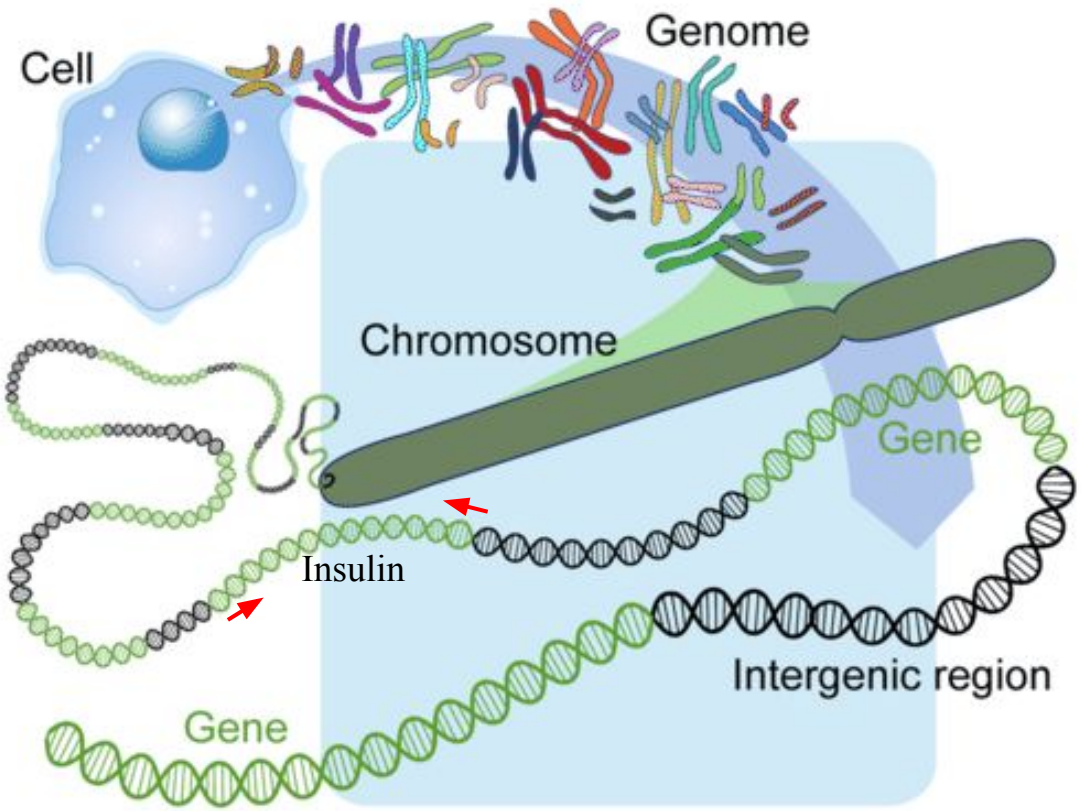
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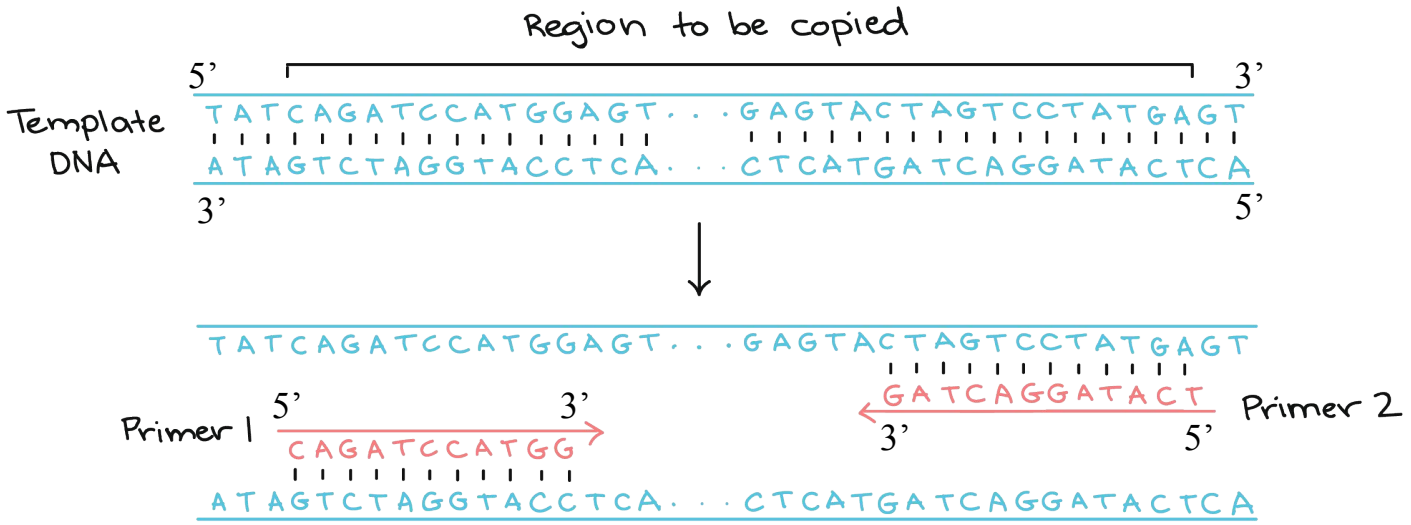
1. PCR (Polymerase Chain Reaction) to amplify specific segment of DNA (gene)
2. Agarose Gel Electrophoresis to separate DNA based on molecular weight
3. DNA Fingerprinting to identify individuals (parent/kids/crime suspects)
4. SDS-PAGE to separate proteins based on molecular weight
5. RT-PCR to quantify DNA/RNA copy numbers
6. Fluorescence Microscopy to visualize cellular processes

Polymerase Chain Reaction (PCR) to amplify specific gene from the genome

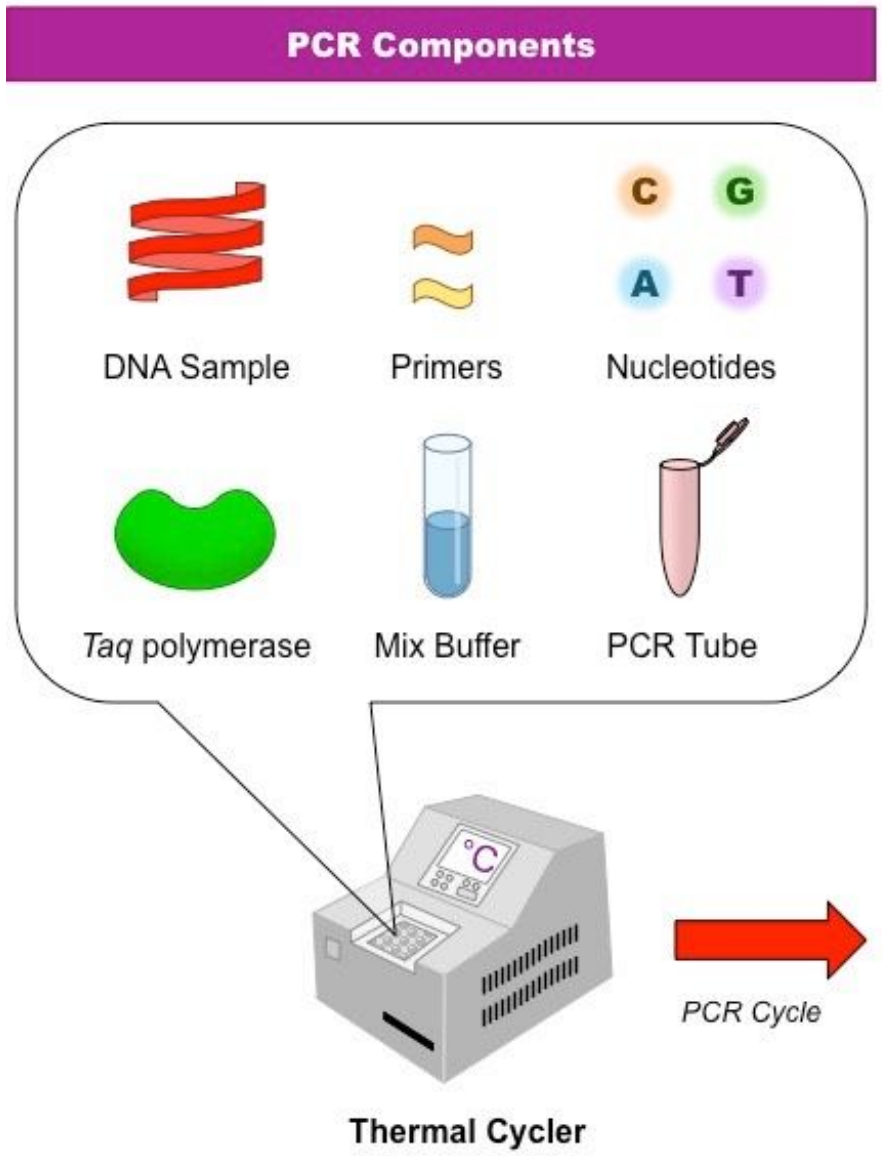
- PCR is used to amplify a segment of DNA (gene). It can make thousands of copies of a gene.

- DNA is synthesized by an enzyme called DNA polymerase. It synthesize DNA in 5' to 3' direction.

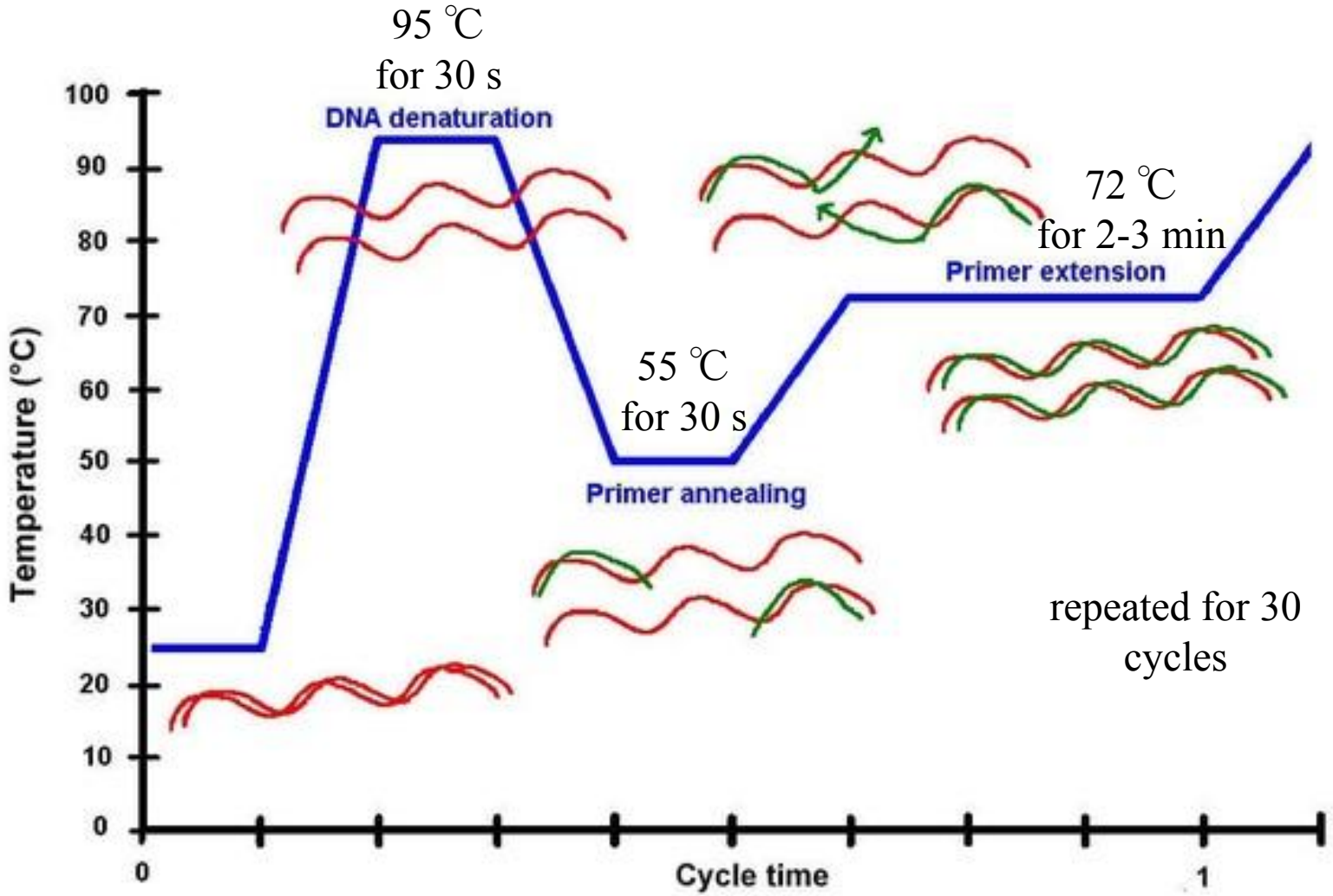




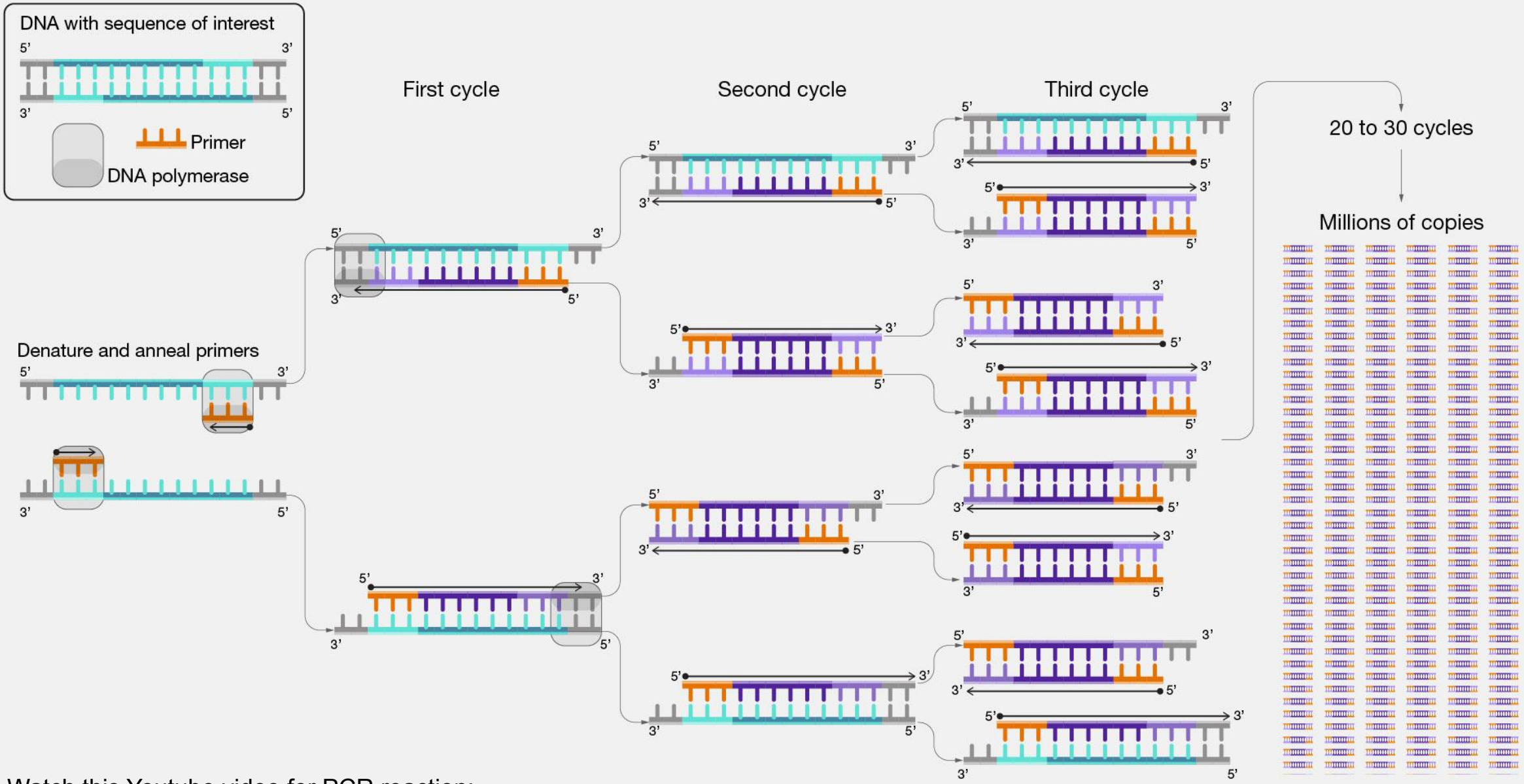
Components of the PCR reaction and amplification cycle



Taq DNA polymerase is thermostable enzyme
Speed of DNA synthesis by Taq DNA polymerase: 2000 bp/min



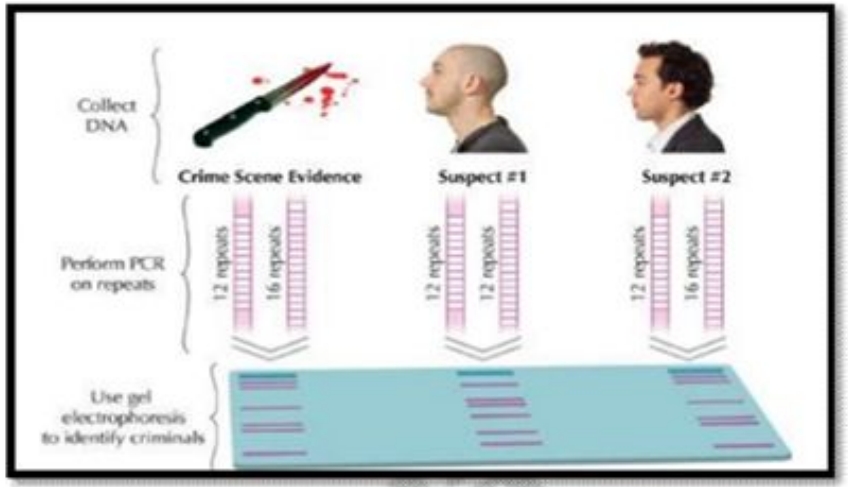
Exponential DNA amplification by PCR



Watch this Youtube video for PCR reaction:
https://www.youtube.com/watch?v=iQsu3Kz9NYo

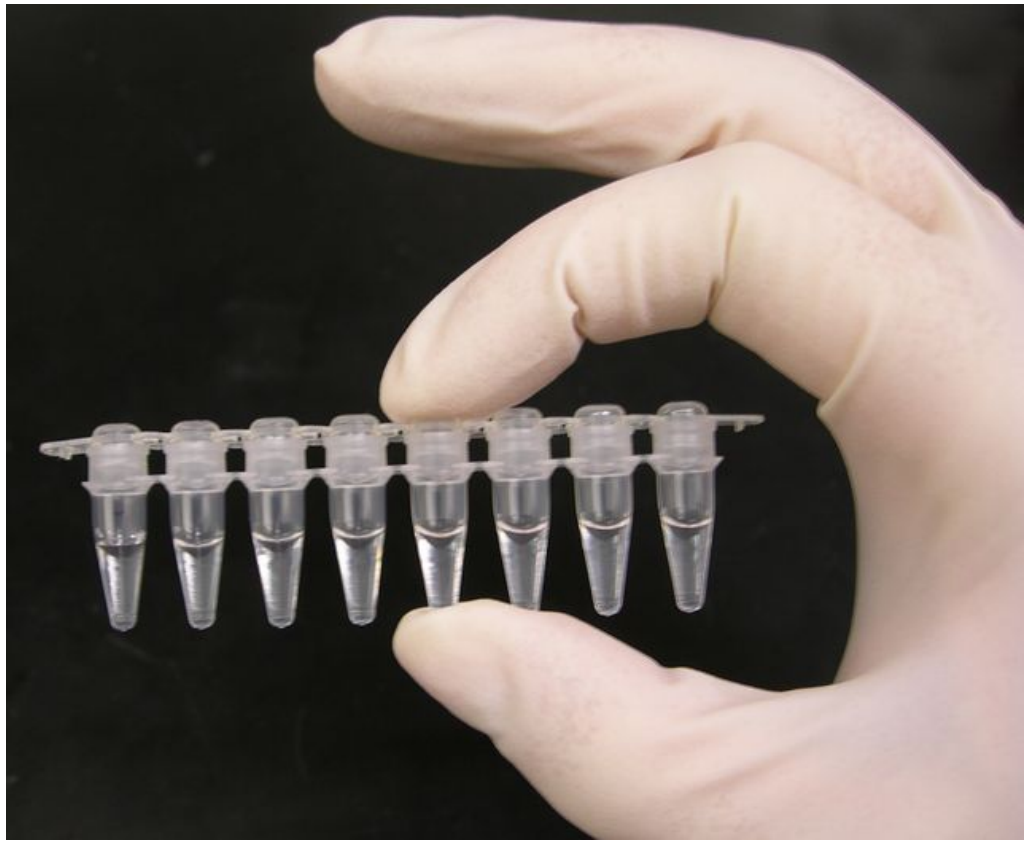
Applications of PCR

- Amplify a specific fragment of DNA (a gene) for cloning purposes.
- Diagnostic applications to detect bacterial or viral infections.
- Forensic DNA analysis to match DNA to criminal suspects. (based on DNA fingerprinting)
- Paternity and maternity testing. (based on DNA fingerprinting)
- DNA fingerprinting act as a genetic barcode to identify individuals.
- Diagnosis of genetic disorders.

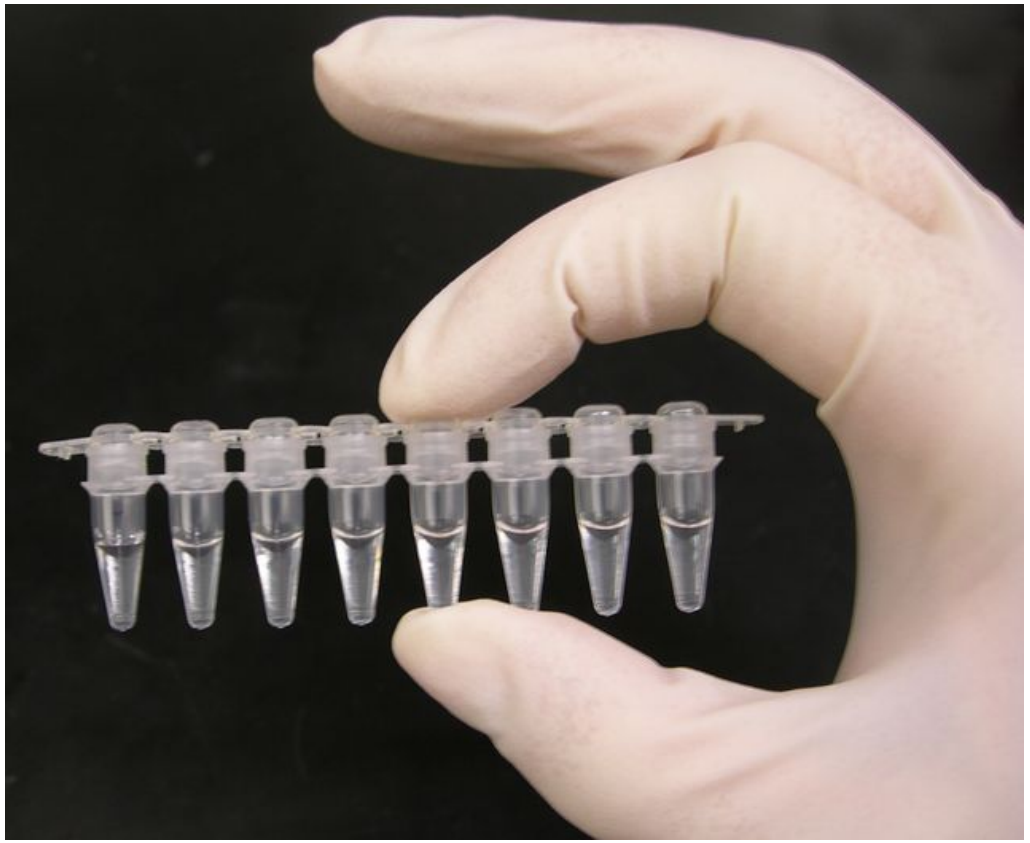


PCR tubes with reaction mixture

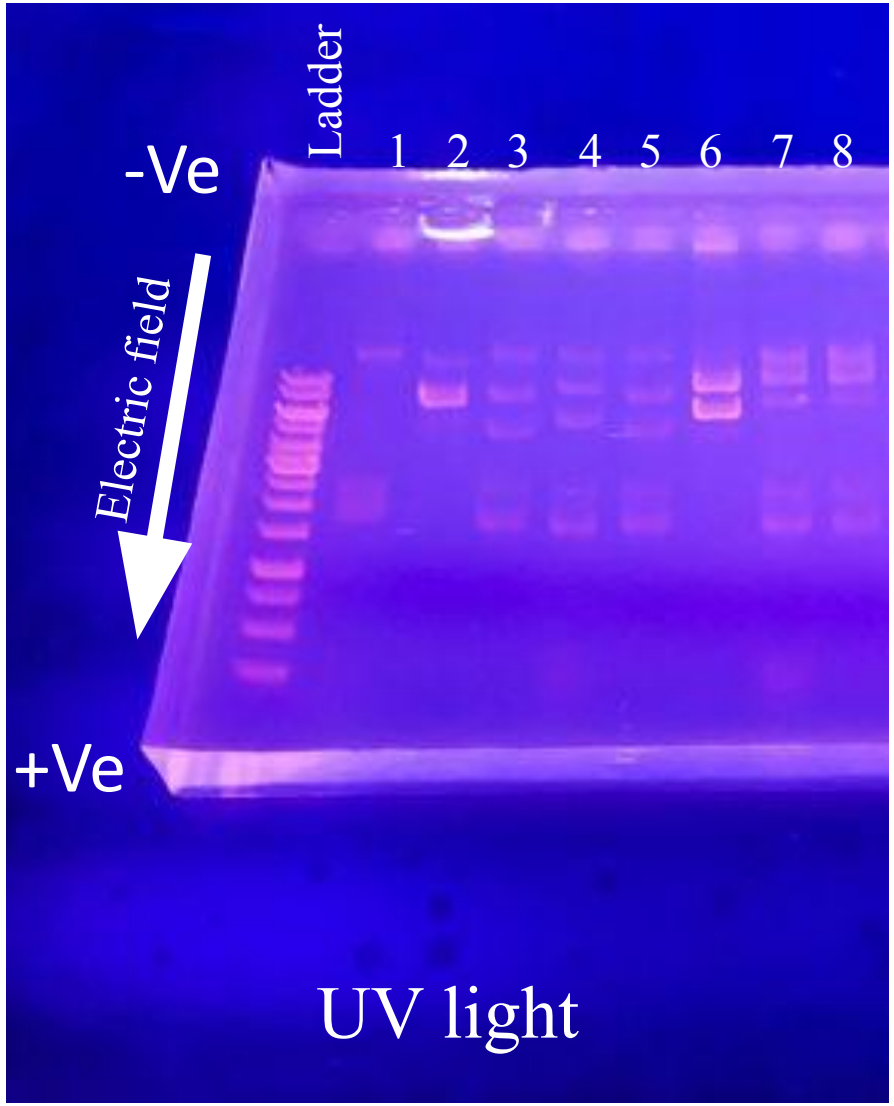
Before DNA amplification/
Before PCR reaction



After DNA amplification/
After PCR reaction



Agarose Gel Electrophoresis to separate and visualize DNA



Samples

large DNA

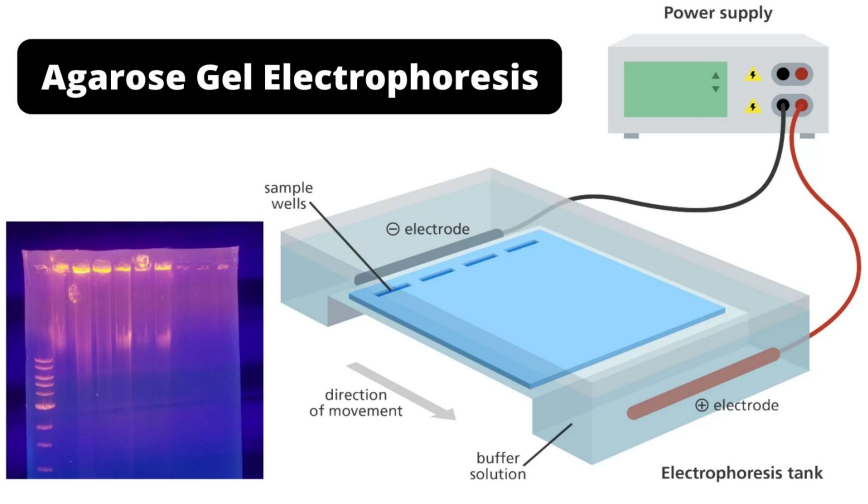
small DNA

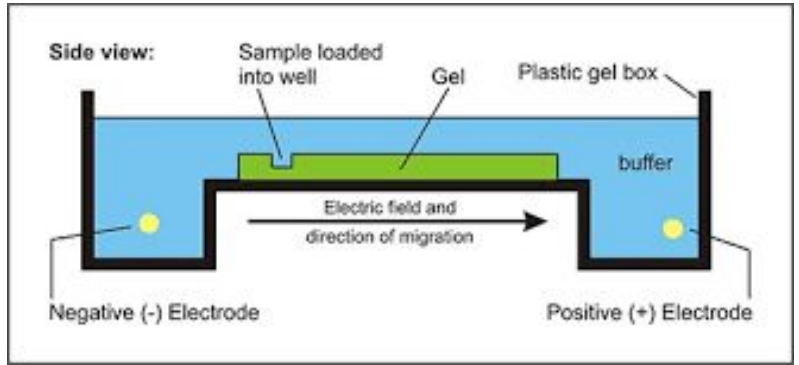
Gelatin gels that we eat

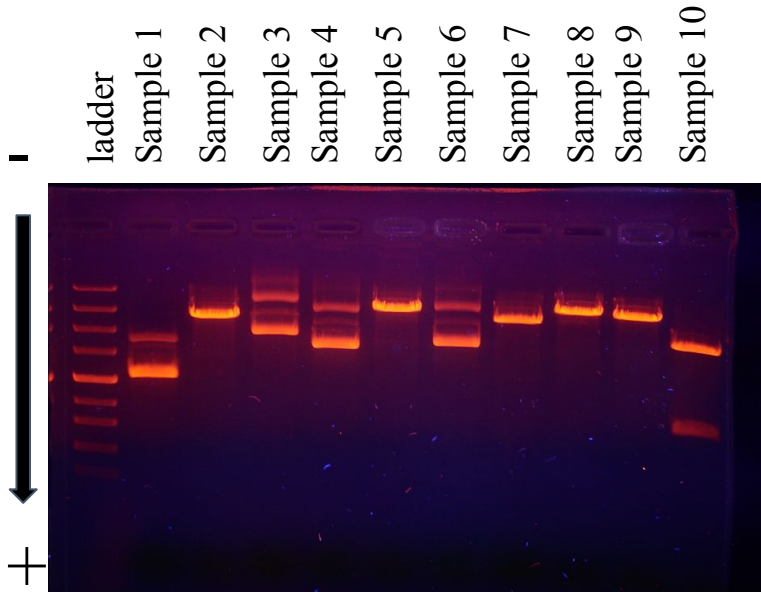


Agarose Gel Electrophoresis to separate DNA

- Agarose gel electrophoresis is a method to separate a mixed population of DNA, based on their sizes (molecular weight).
- Agarose gel is a 3D matrix, in which the mixture of DNA is loaded and the electric field is applied.
- DNA migrates from negative to positive electrode and the distance it migrates depends on the size of the DNA. Small DNA molecules (e.g. 300 bp) moves faster, compared to large DNA molecules (i.e. 1000 bp).
- The agarose gel is stained with ethidium bromide (EtBr) and observed under the Ultraviolet light to see the bands of DNA.
- DNA molecular weight marker (DNA ladder) is used as a reference of molecular weight.







DNA Fingerprinting/RFLP (Restriction Fragment Length Polymorphism)

- DNA Fingerprinting is a method in which specific DNA sequences are amplified from the whole genome by PCR, followed by restriction digestion and agarose gel electrophoresis to visualize the fragmentation patterns to identify individuals.

- DNA Fingerprinting is also called Restriction Fragment Length Polymorphisms (RFLPs) Analysis and Random Amplified Polymorphic DNA (RAPD).

- For DNA Fingerprinting, microsatellite DNA is amplified using PCR from the genome. The number of repeated segments within a microsatellite sequence often varies among people, which makes them **useful as polymorphic markers for studying inheritance patterns in families or for creating a DNA fingerprint from crime scene samples**.

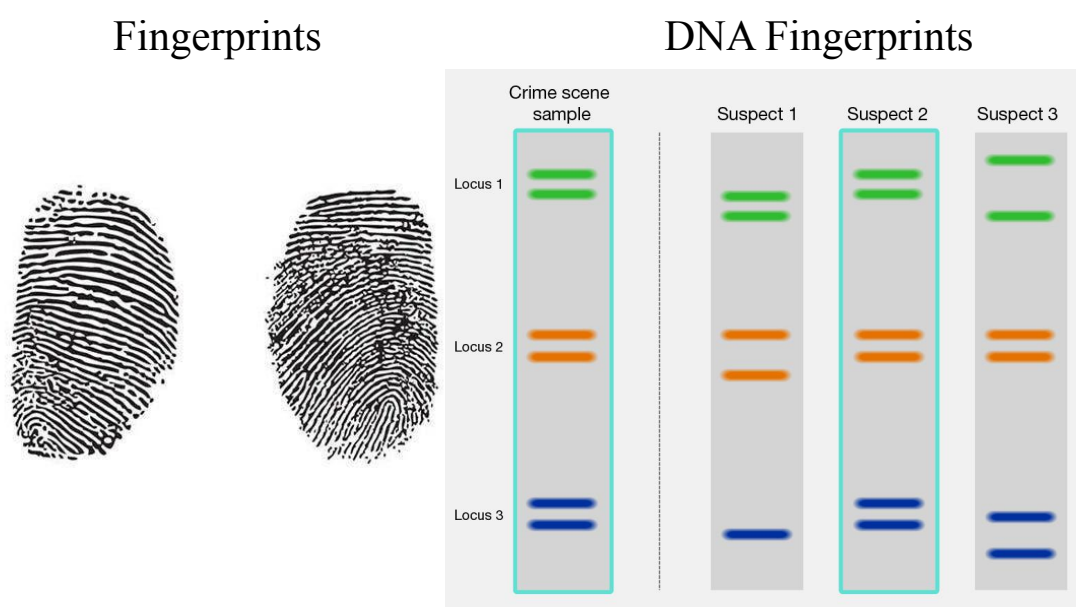
- DNA fingerprinting has four main steps:

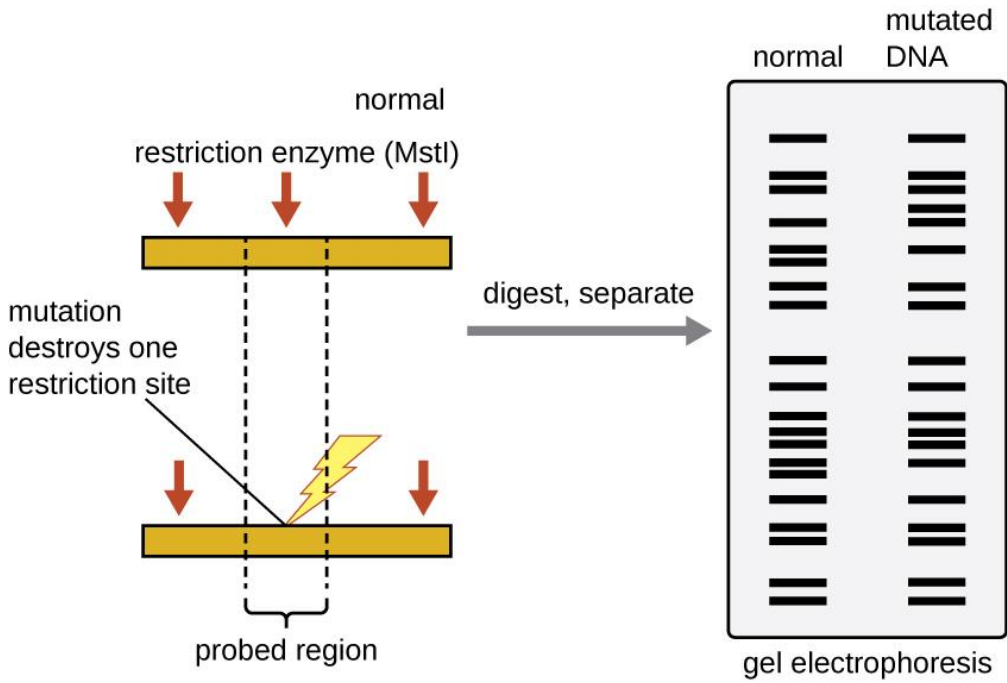
  Step 1: DNA extraction from cells.

  Step 2: PCR to amplify the extracted DNA.

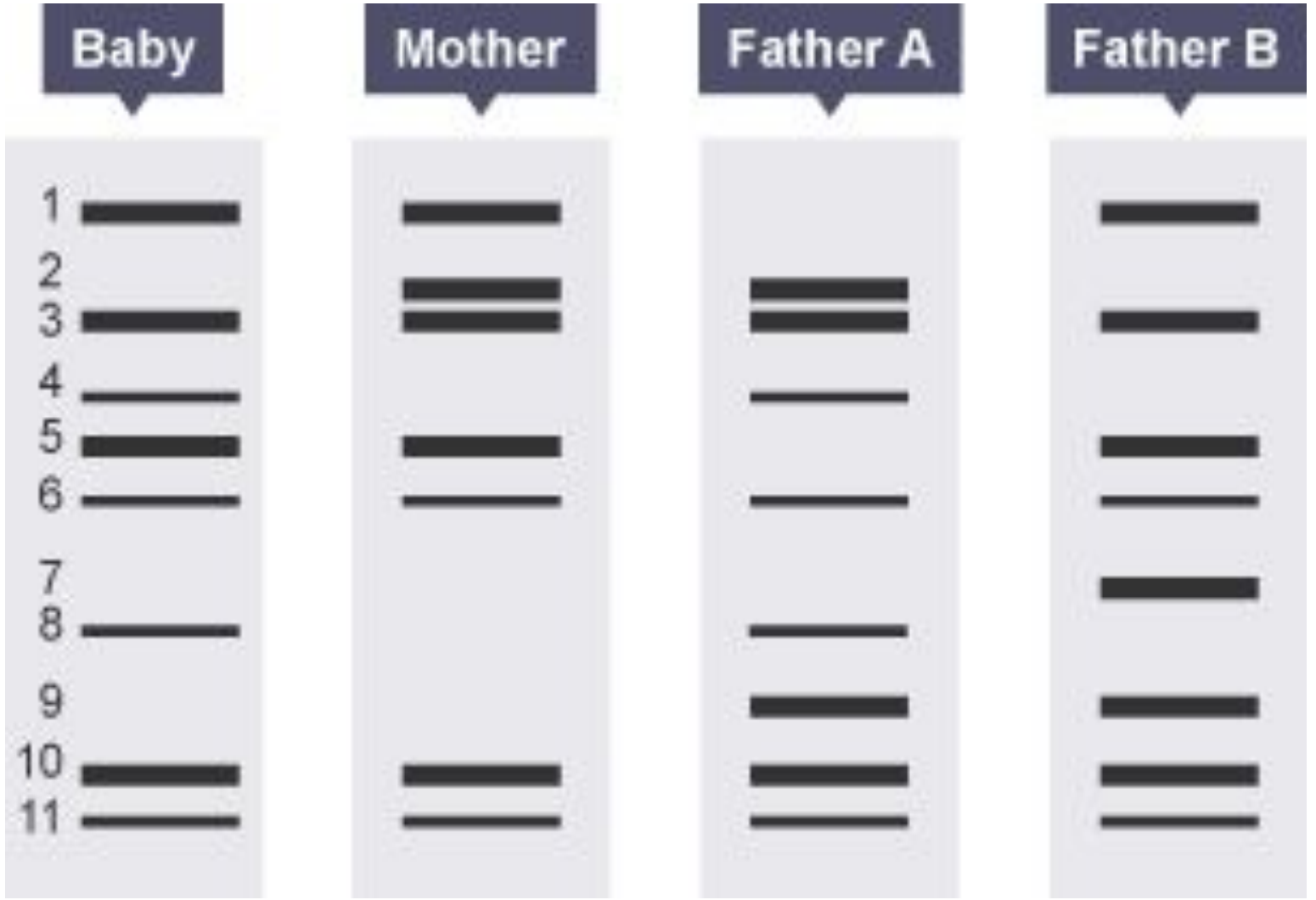
  Step 3: Restriction enzyme treatment to cut the DNA at specific sequences, resulting in fragments that are unique in size to each person.

  Step 4: Gel electrophoresis to separate these DNA fragments by size for comparison.

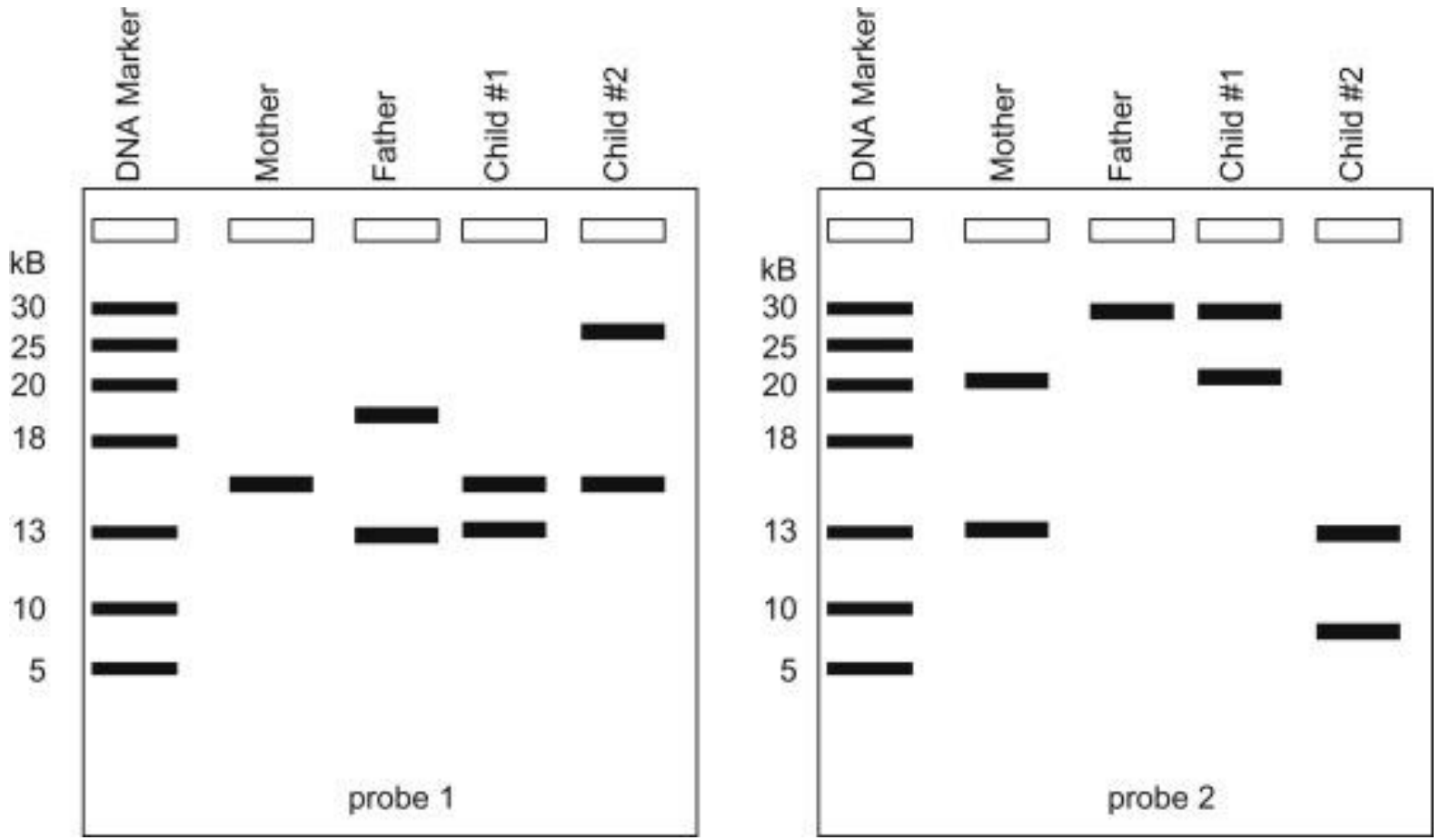




Identify the father of this baby from this DNA Fingerprinting Gel



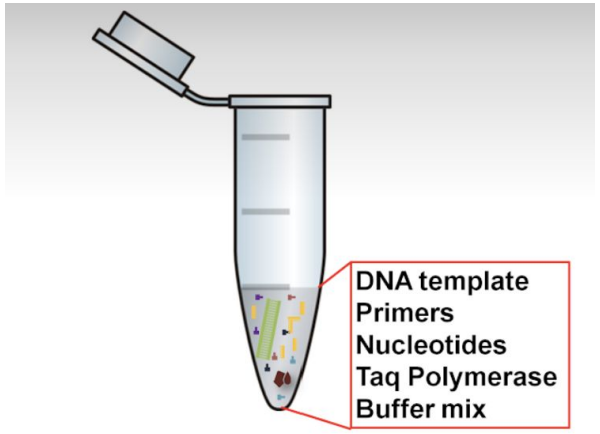
Identify the real child of these mother and father from this DNA Fingerprinting Gel



Real-Time PCR (RT-PCR) or Quantitative PCR (q-PCR)

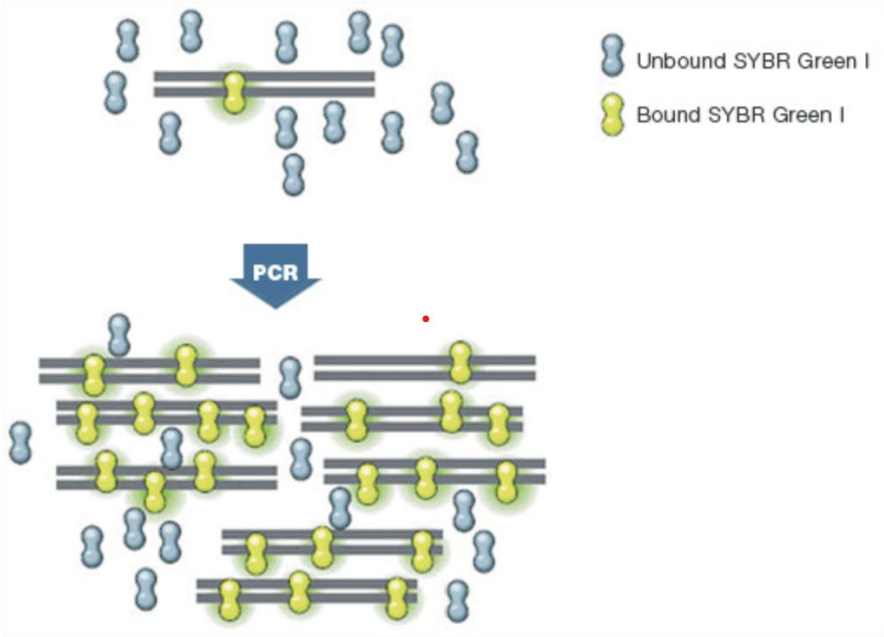
RT-PCR or qPCR: It is performed in the same way as PCR, however, a fluorescent dye is added to the tube that gives fluorescence proportional to the amount of double stranded DNA in the tube. So, the quantify of the double stranded DNA is measure in real-time while the PCR is going on.

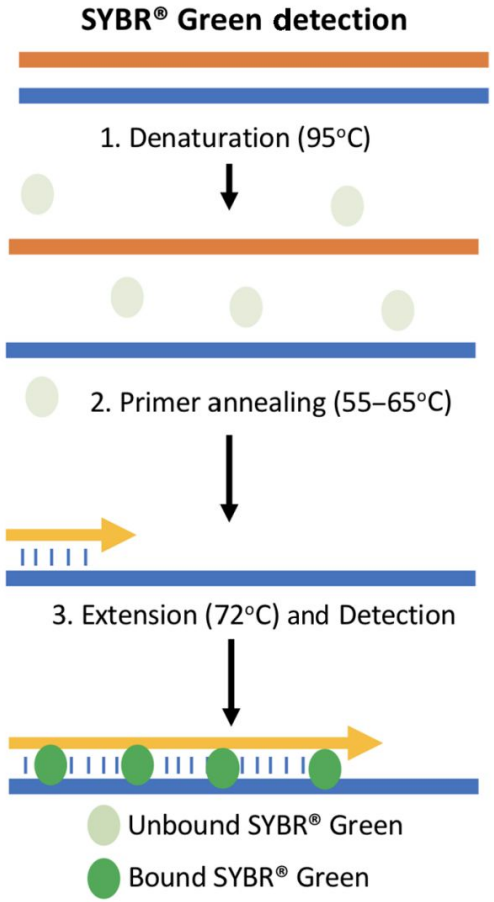
This technique is used to quantify the number of DNA molecules present in the tube/sample.



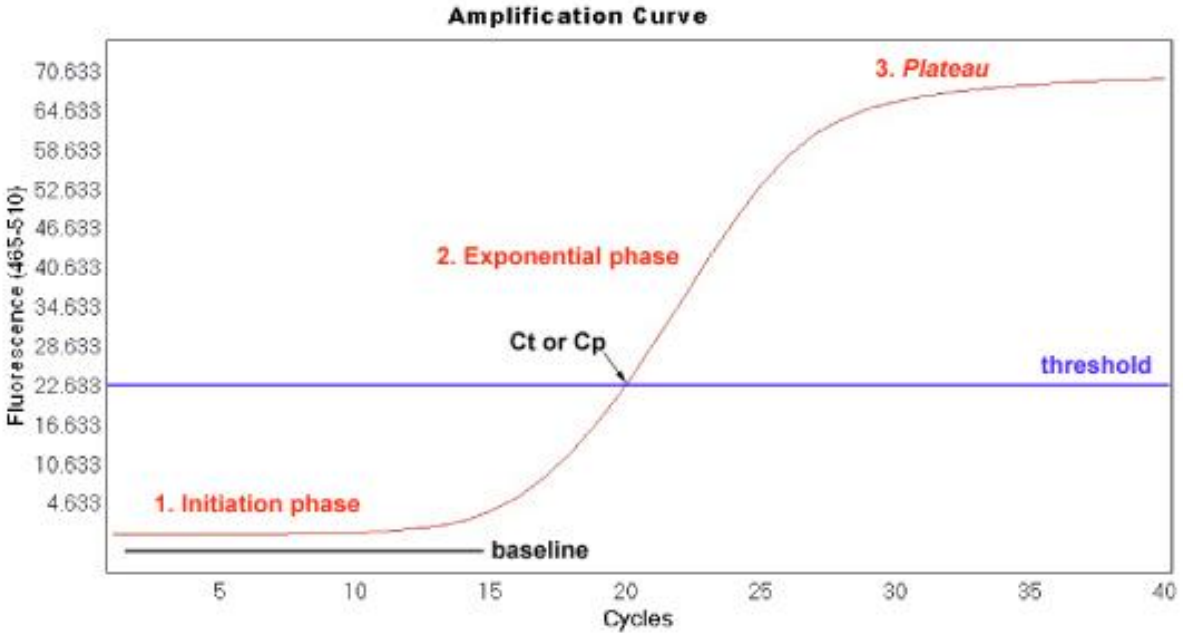
+

Syber green

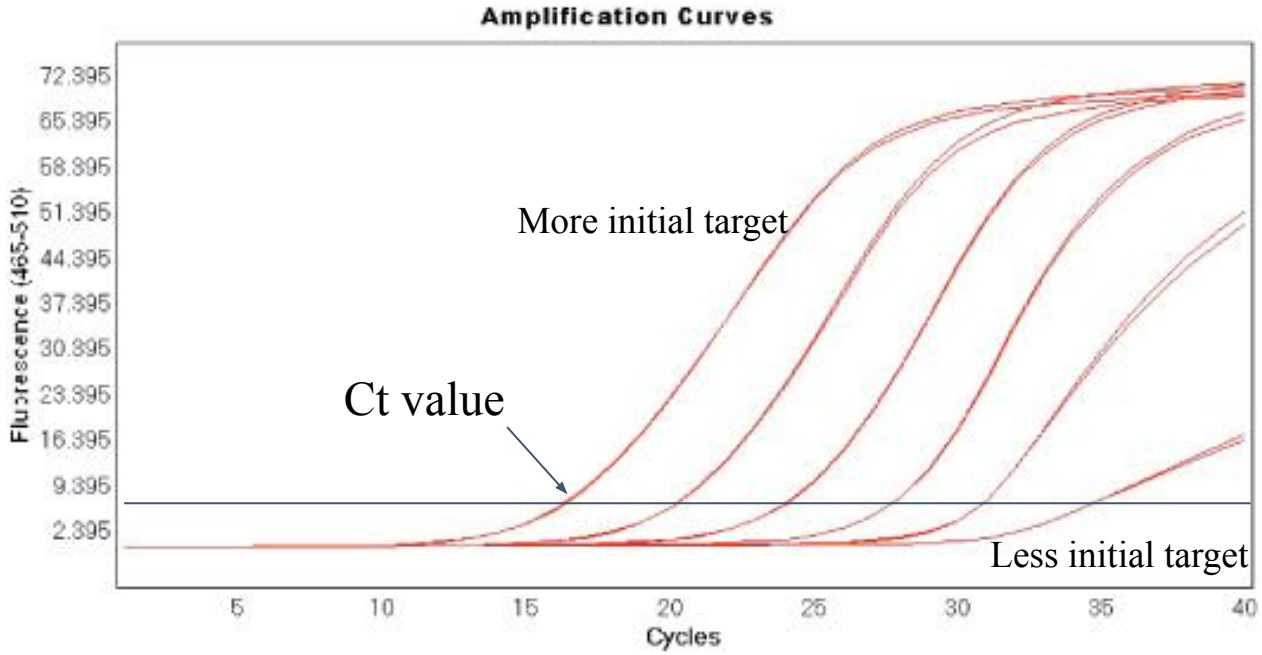




Phases of a RT-PCR amplification curve

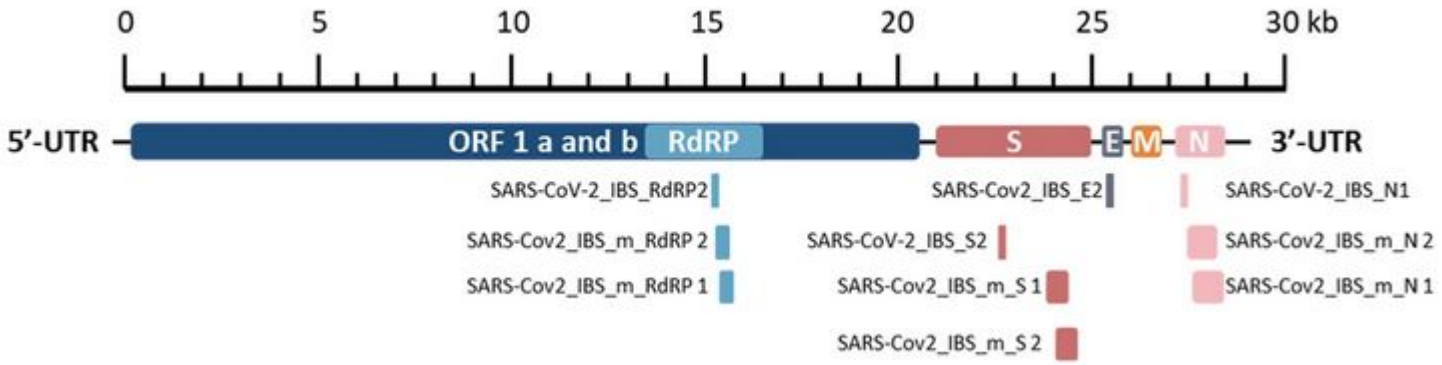


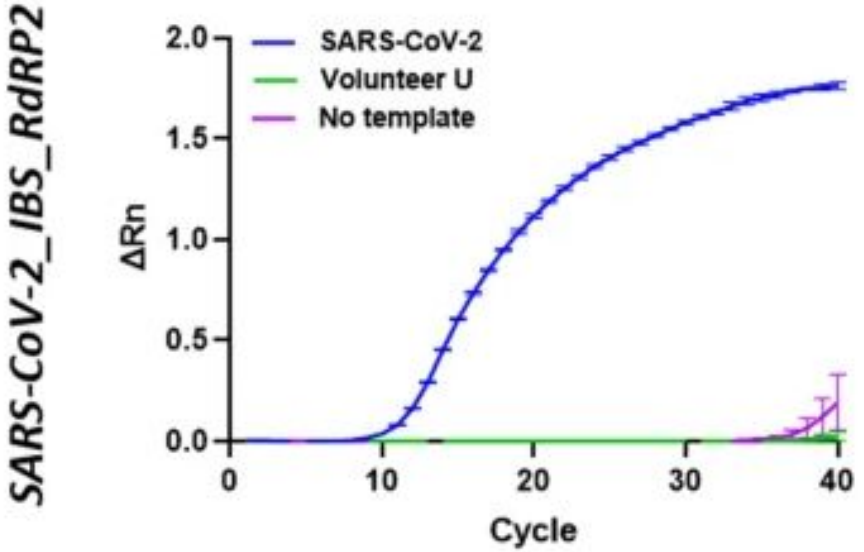
Amplification curve obtained from serial dilutions of a target DNA

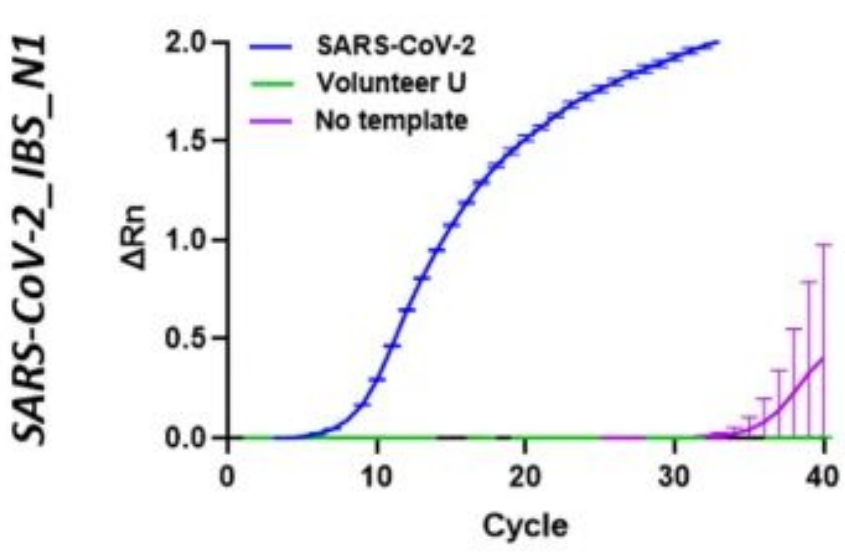


Ct value (Critical Threshold Value): It is the number of PCR cycle at which the fluorescence intensity of SyBR green dye became so high that the detector can see green fluorescence in the tube.

Genome of Covid19 virus



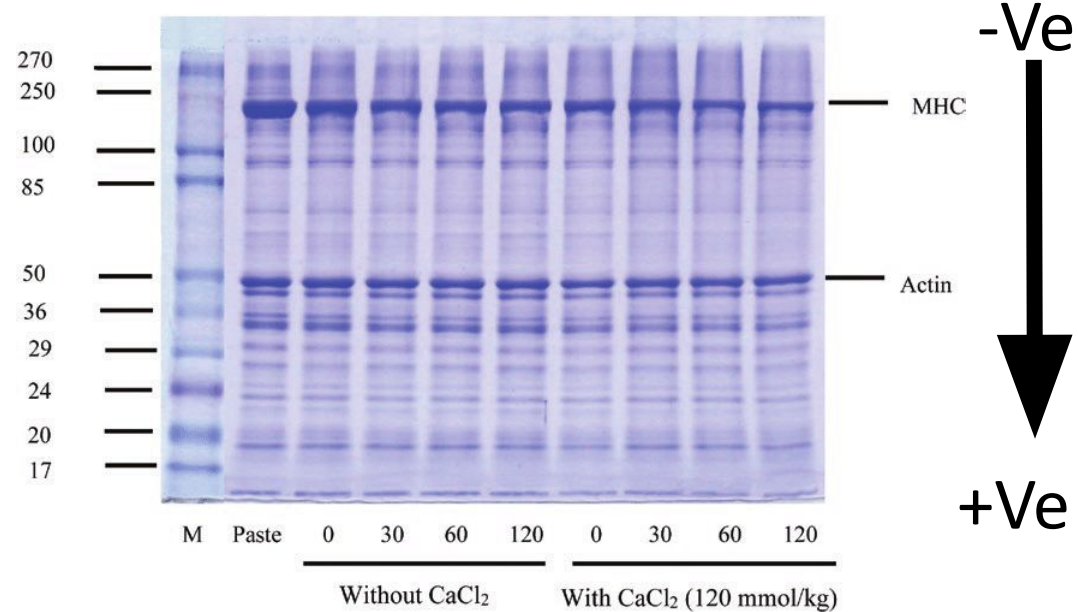




Polyacrylamide Gel Electrophoresis to separate proteins (SDS-PAGE)

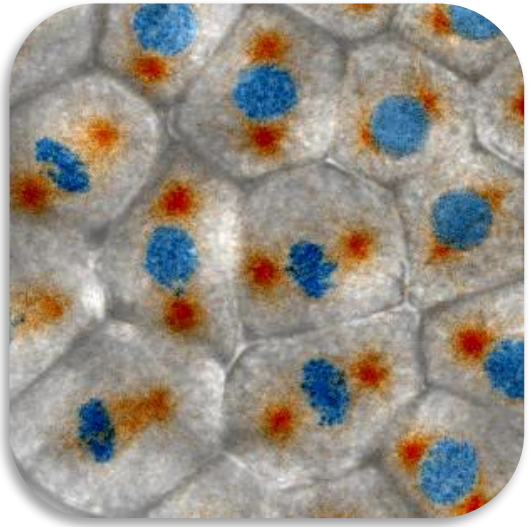
- SDS-PAGE or Sodium Dodecyl Sulphate-PolyAcrylamide Gel Electrophoresis is a method used for separation of proteins based on their molecular weight.
- SDS imparts negative charge to the proteins.
- Gel is made up of Polyacrylamide and protein mixture is loaded on the top.
- Electric field is applied, so the proteins migrate from negative to positive direction. The smaller proteins migrate faster compared to the larger proteins.



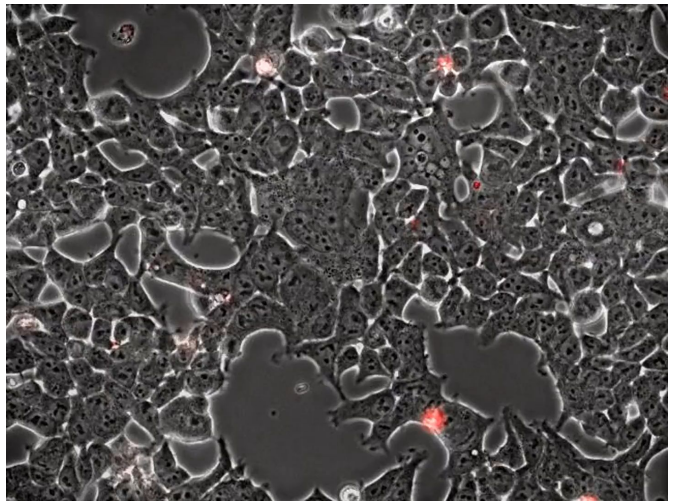


Fluorescence Microscopy to visualize the cellular processes

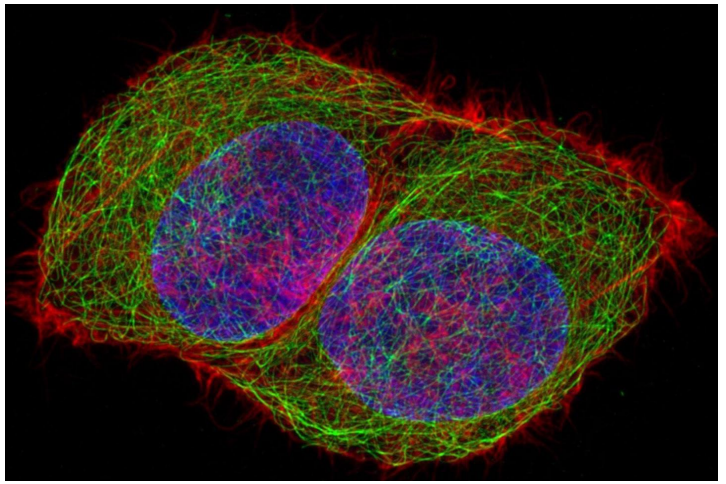
Seeing is believing



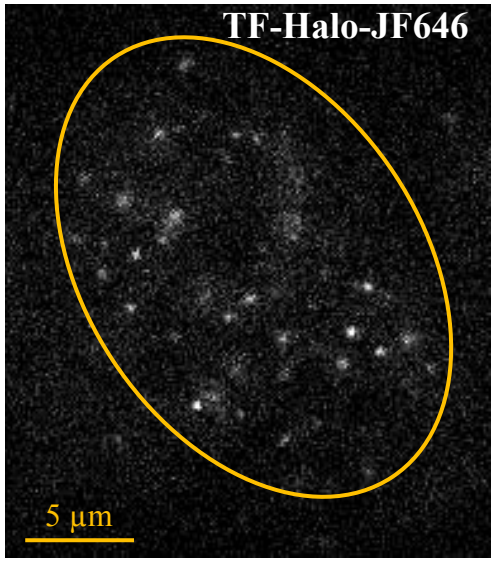
How cell divides and chromosomes Segregate?



How SARS-CoV-2 causes cell fusion and death in bat brain cells?



HeLa cells stained for DNA (blue), microtubules (green) and F-actin (red)



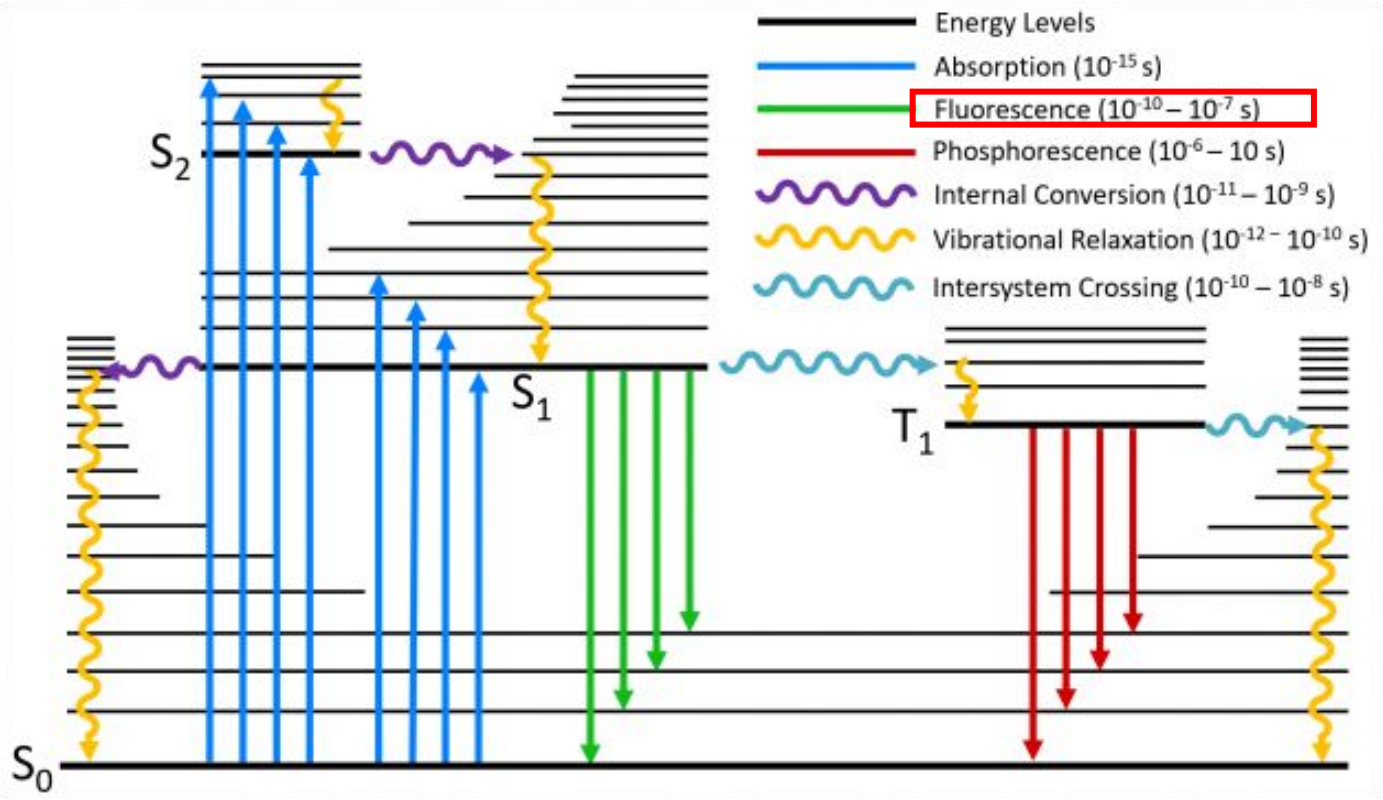
Single-Molecule Imaging of transcription factor (TF) in human cell nucleus

Mouse retina, confocal 3D microscopy: https://youtu.be/Jo7MvM_IXa8
3D imaging of collagen fibre network: https://youtu.be/pof6ywsTK6Y
Kidney cells mitosis: https://youtu.be/ed6OkXi6M5Q
Zebrafish embryonic development at single-cell resolution: https://youtu.be/RQ6vkDr_Dec

What is fluorescence?

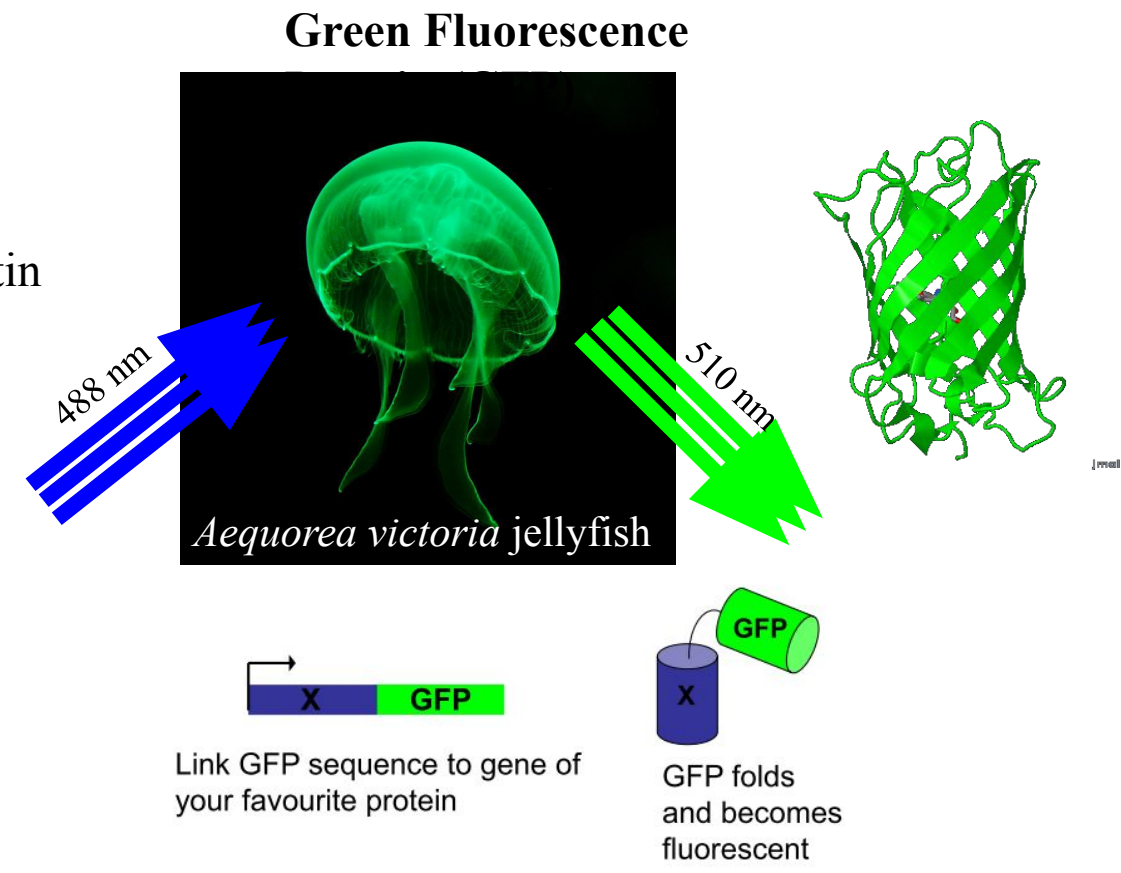
- Fluorescence is the emission of light by a molecule that has absorbed light.
- Fluorescence has higher wavelength than absorbed light.

Jablonski diagram

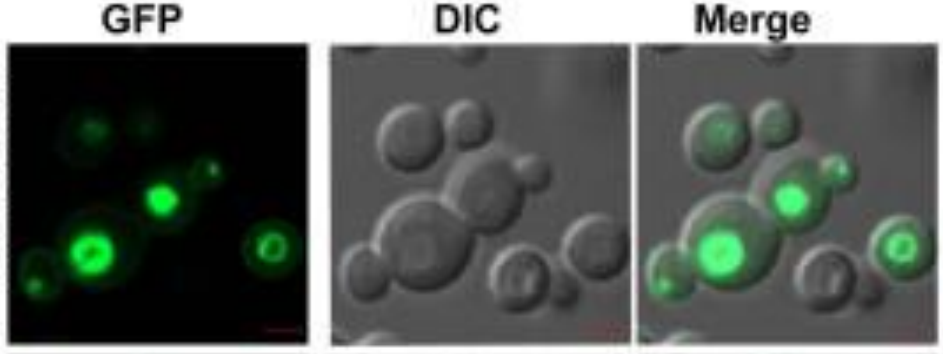


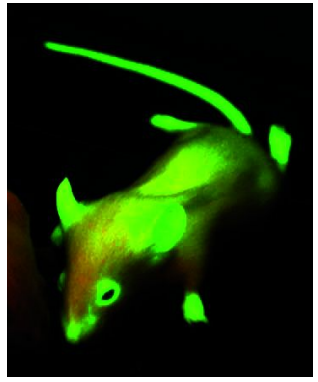
Discovery of Green Fluorescent Protein (GFP) leads to revolution in fluorescence microscopy

- The prospects of fluorescence microscopy changed dramatically with the discovery of fluorescent proteins in the 1950s.
- GFP was first discovered in the jellyfish *Aequorea victoria*.
- Nobel prize awarded to Roger Tsien, Osamu Shimomura and Martin Chalfie in 2008 for their discovery of GFP.
- Fluorescence due to three amino acids at positions S65, T66, G67.
- Major excitation peak at 488 nm (blue light)
  Emission peak is at 510 nm (green light)
  Molecular weight ~27 kD
  Amino Acids: 238



Yeast cells expressing GFP labelled protein

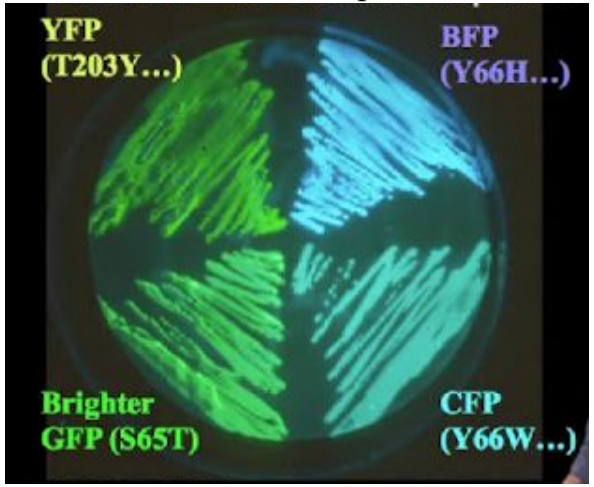


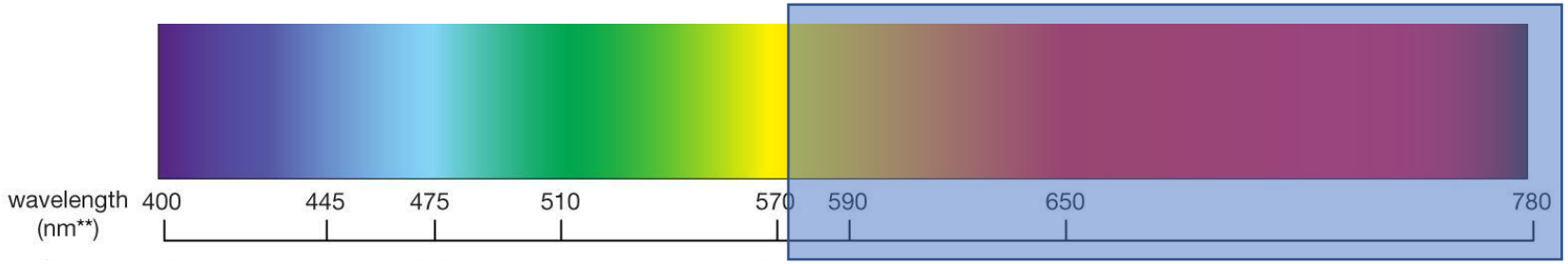


Protein engineering to change the spectral and physical properties of GFP

- **EGFP:** S65T mutation leads to five times brighter GFP than original GFP. Ex: 488 nm, Em: 510 nm
- **Sapphire:** T203I mutation leads to change in excitation and emission. Ex: 399 nm, Em: 511 nm.
- **BFP** (Blue Fluroescent Protein): Y66H, Ex: 380 nm, Em: 448 nm
- **CFP** (Cyan Fluorescent Protein): Y66W, Ex: 433/445 nm, Em: 475/503 nm
- **YFP** (Yellow Fluroescent Protein): T203Y, Ex: 514 nm, Em: 527 nm
- **GFP** and its variants covered the entire lower wavelength spectra of the visible light.
- **Red FPs are missing.**

Colors of GFP mutants expressed in *E. coli*

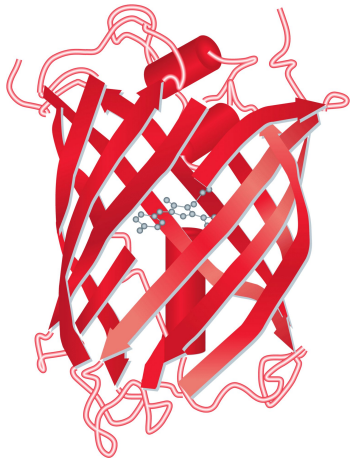




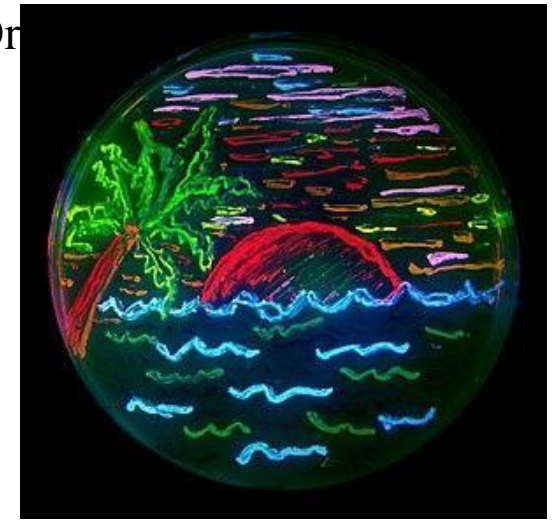
https://www.leica-microsystems.com/science-lab/fluorescent-proteins-introduction-and-photo-spectral-characteristics/

Amino Acid	3-Letter Code	1-Letter Code
Alanine	Ala	A
Cysteine	Cys	C
Aspartic acid or aspartate	Asp	D
Glutamic acid or glutamate	Glu	E
Phenylalanine	Phe	F
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Lysine	Lys	K
Leucine	Leu	L
Methionine	Met	M
Asparagine	Asn	N
Proline	Pro	P
Glutamine	Gln	Q
Arginine	Arg	R
Serine	Ser	S
Threonine	Thr	T
Valine	Val	V
Tryptophan	Trp	W
Tyrosine	Tyr	Y

Fluorescent Proteins from Anthozoa (Red spectrum)



- Russian scientist Sergey Kukyanov discovered Red Fluorescent Protein (RFP) in **Anthozoa**.
- **DsRed:** Ex: 558, Em: 583.
- **DsRed2:** Faster maturation and reduced oligomerization.
- **mRFP1:** first monomeric RFP
- Further genetic manipulations to mRFP1 leads to six mFruit FPs (mHoneydew, mBanana, mOr mTangerine, mStrawberry, mCherry).
- **mCherry** is widely used: Em: 610 nm, brightness 50% of EGFP
- **mPlum:** Em: 649 nm (Red FP with highest emission wavelength)
- **mKate2:** Em: 635 nm (Brightest Red FP)



The diversity of genetic mutations is illustrated by living bacteria expressing 8 different colors of fluorescent proteins (derived from GFP and dsRed).

Fluorescent Proteins in Bio-Imaging

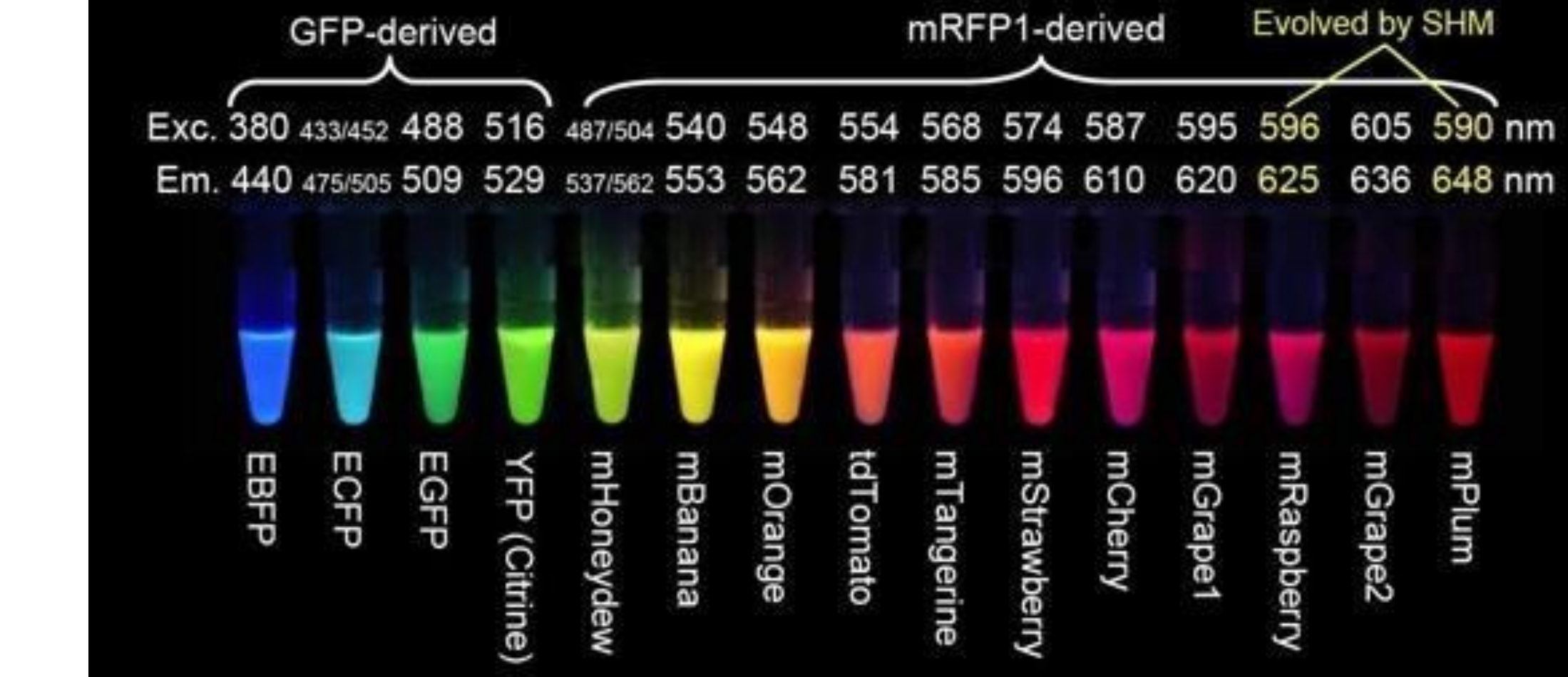
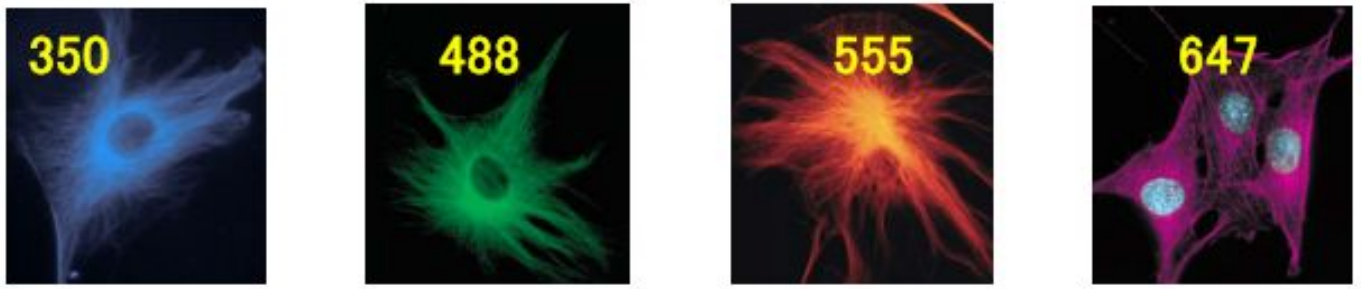


Image showing widely used Fluorescent Proteins and their peak emission / absorption wavelengths.
Image Courtesy: Roger Y Tsien Nobel Lecture, 2008

Multiplexing– four main colours

Excitation wavelengths:



Emission wavelengths:

Blue | green | orange/red | far red

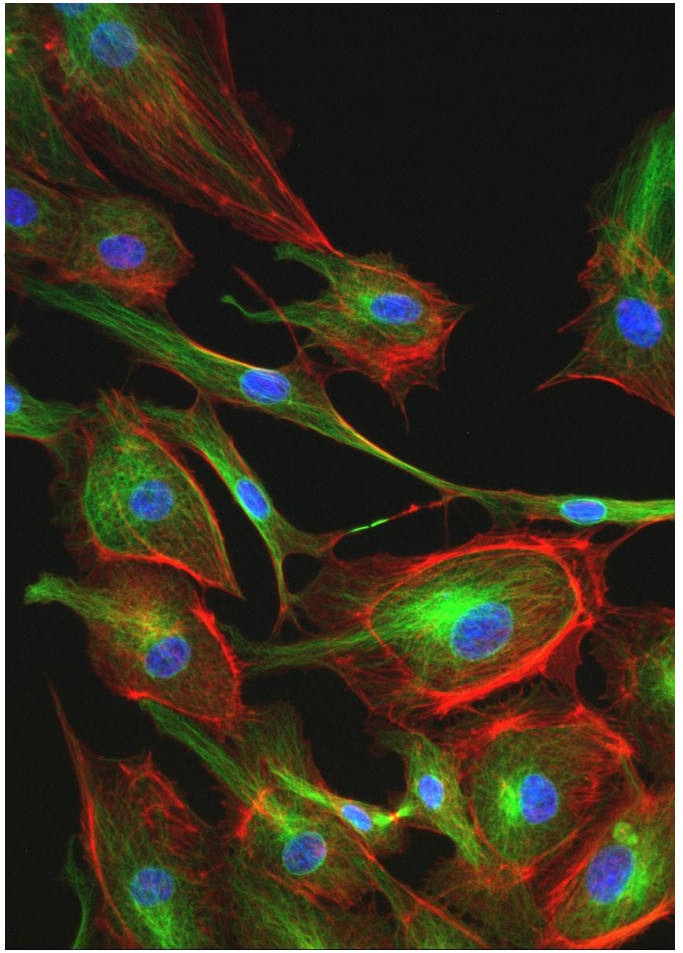
350 400 450 500 550 600 650

DAPI/UV	FITC	TRITC	FAR RED
Alexa Fluor® 350 Coumarin, AMCA	Alexa Fluor® 488 Fluorescein (FITC) Cy2	Alexa Fluor® 555 Rhodamine, TAMRA, TRITC Cy3	Alexa Fluor® 647 Cy5, APC

Alexa Fluor® 594
Texas Red, Cy3.5

Colour Selection ◆ Brightness ◆ Photostability

Blue: DAPI stained cell nucleus
Green: GFP labelled tubulin protein
Red: RFP labelled actin protein



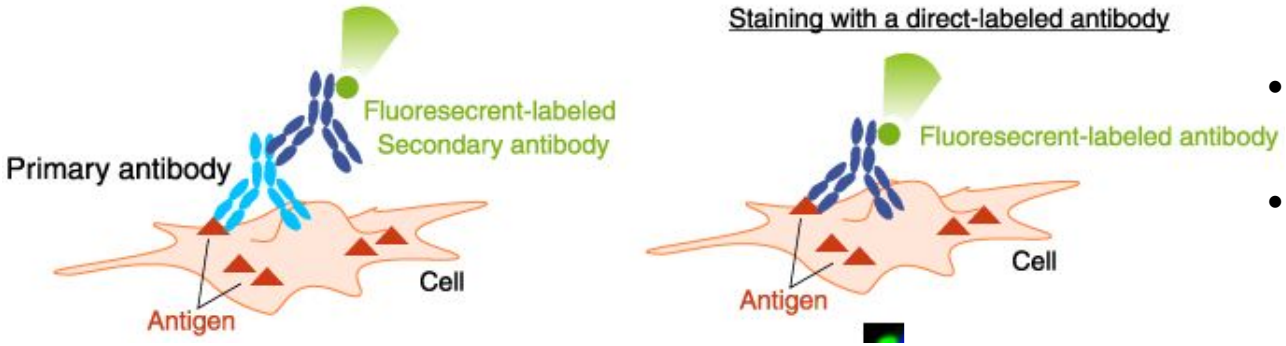
Strategies for labelling proteins with fluorescence

- Genetic encoding: C-terminal or N-terminal fusion of a gene of interest with a gene of fluorescent protein.



- Pros: Specificity of labeling, essential for live cell imaging
- Cons: Time required for genetic engineering

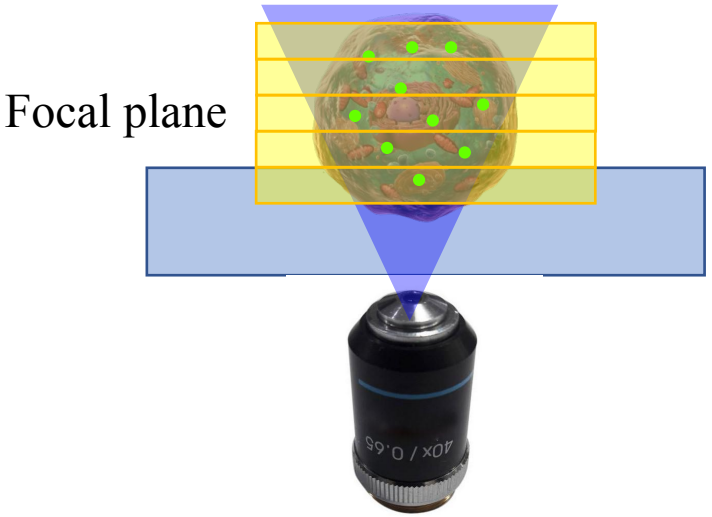
- Antibody based detection: (Immunofluorescence, immunohistochemistry)



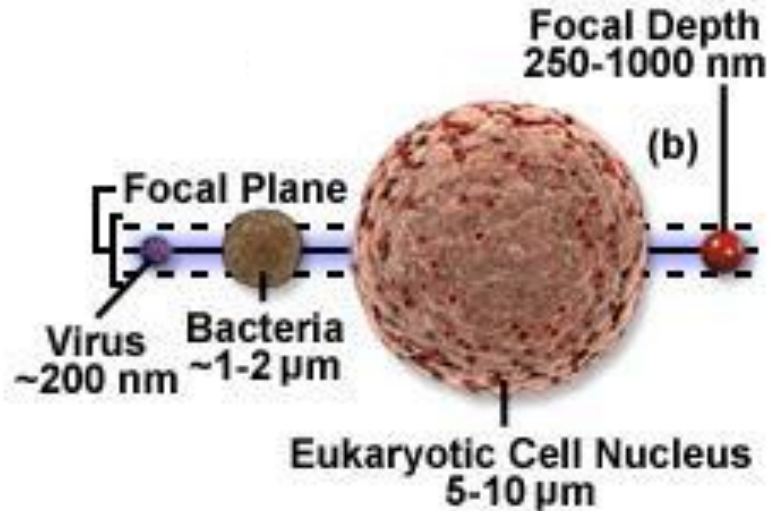
- Pros: No genetic engineering is required if antibody is available commercially. So, less time consuming.
- Cons: suitable for fixed cells (not for live cells)
  Non-specific binding may occur.

A cell is a 3D structure, so it should be imaged in 3D

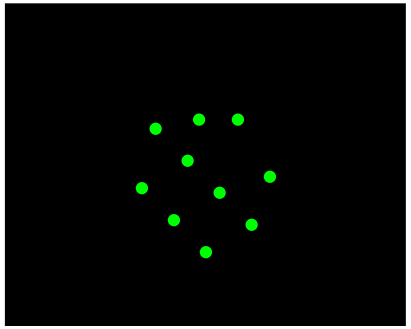
Epifluorescence
(wide field illumination)



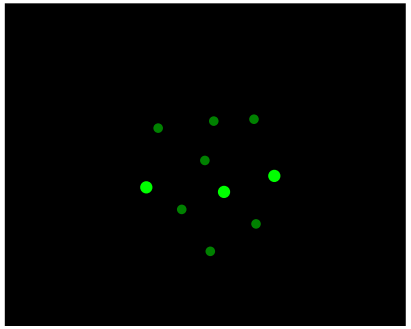
Z-stacking
Z steps= 5



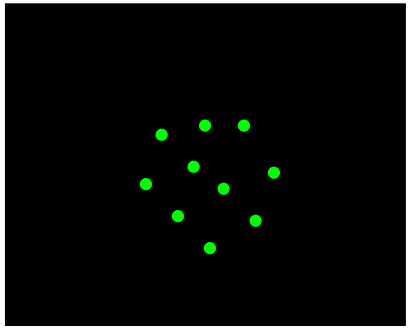
Ideal Image



Actual 2D Image



MIP 3D Image



MIP=Maximum Intensity Projection

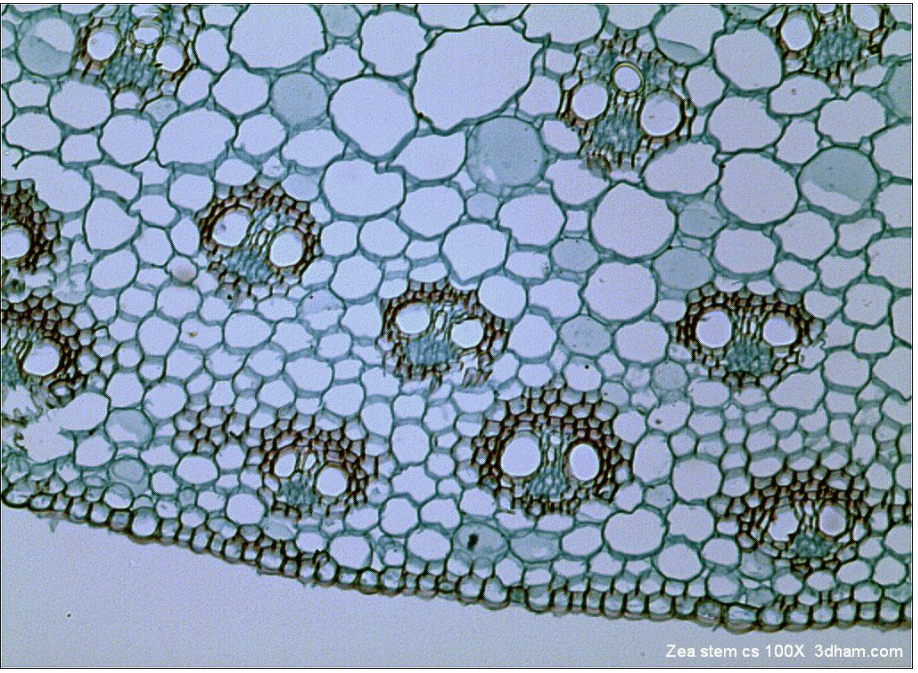
Further reading

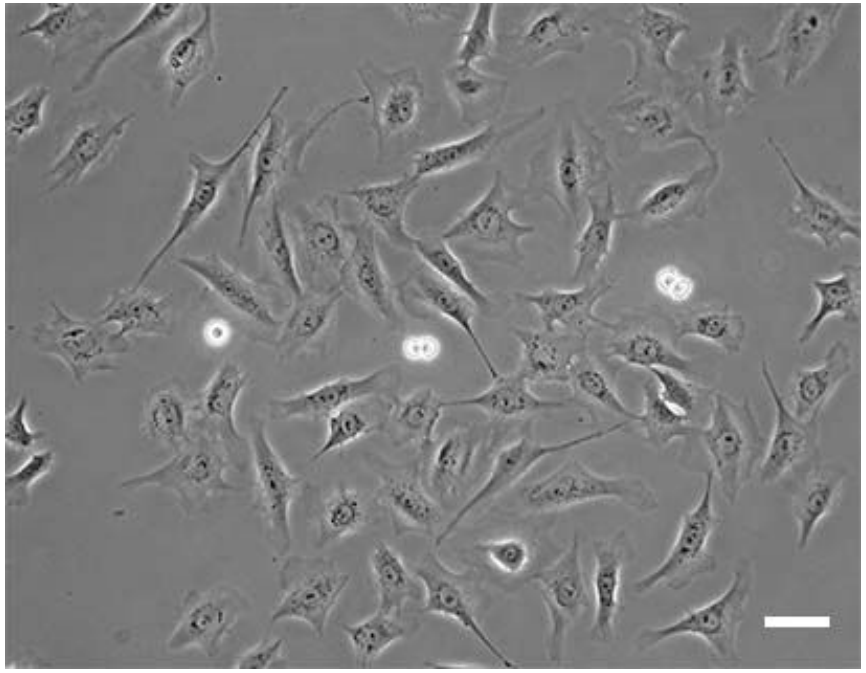
- Book: Fundamentals of Light Microscopy and Electronic Imaging by Douglas Murphy and Michael Davidson, 2nd edition, Wiley-Blackwell publisher, **eBook ISBN** 9781118382912
- Book: Fluorescence Microscopy: From Principles to Biological Applications by Ulrich Kubitscheck, 2nd edition, John Wiley & Sons publisher, **eBook ISBN** 9783527687725

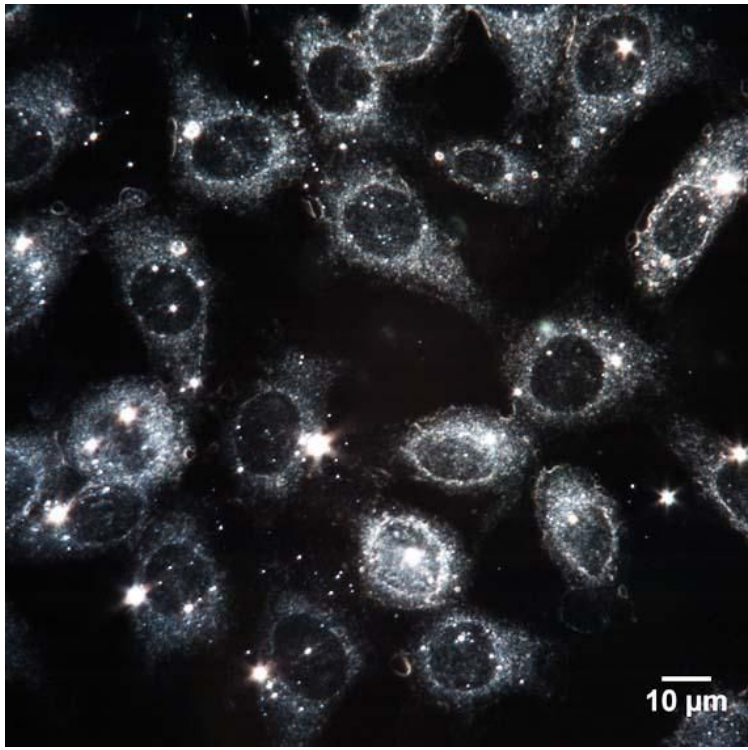
- e-resource: https://www.microscopyu.com/
- e-resource: https://www.ibiology.org/online-biology-courses/microscopy-series/
- e-resource: http://zeiss-campus.magnet.fsu.edu/index.html
- e-resource: https://www.leica-microsystems.com/science-lab/topics/basics-in-microscopy/
- e-resource: https://www.olympus-lifescience.com/en/microscope-resource/
- e-resource: https://imagej.nih.gov/ij/docs/examples/index.html
- e-resource: https://www.youtube.com/playlist?list=PL5ESQNfM5lc7SAMstEu082ivW4BDMvd0U

Slide no. 29 onwards are not a part of your syllabus for the exam on 17/11/2025.

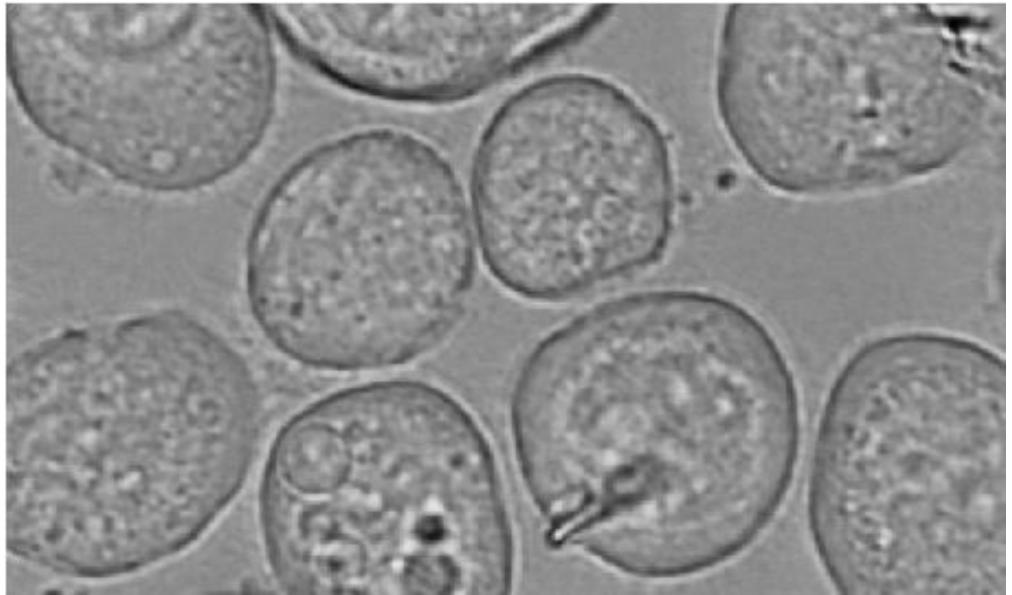
Bright field / Dark Field / Phase Contrast Microscopy













Anatomy of the Fluorescence Microscope

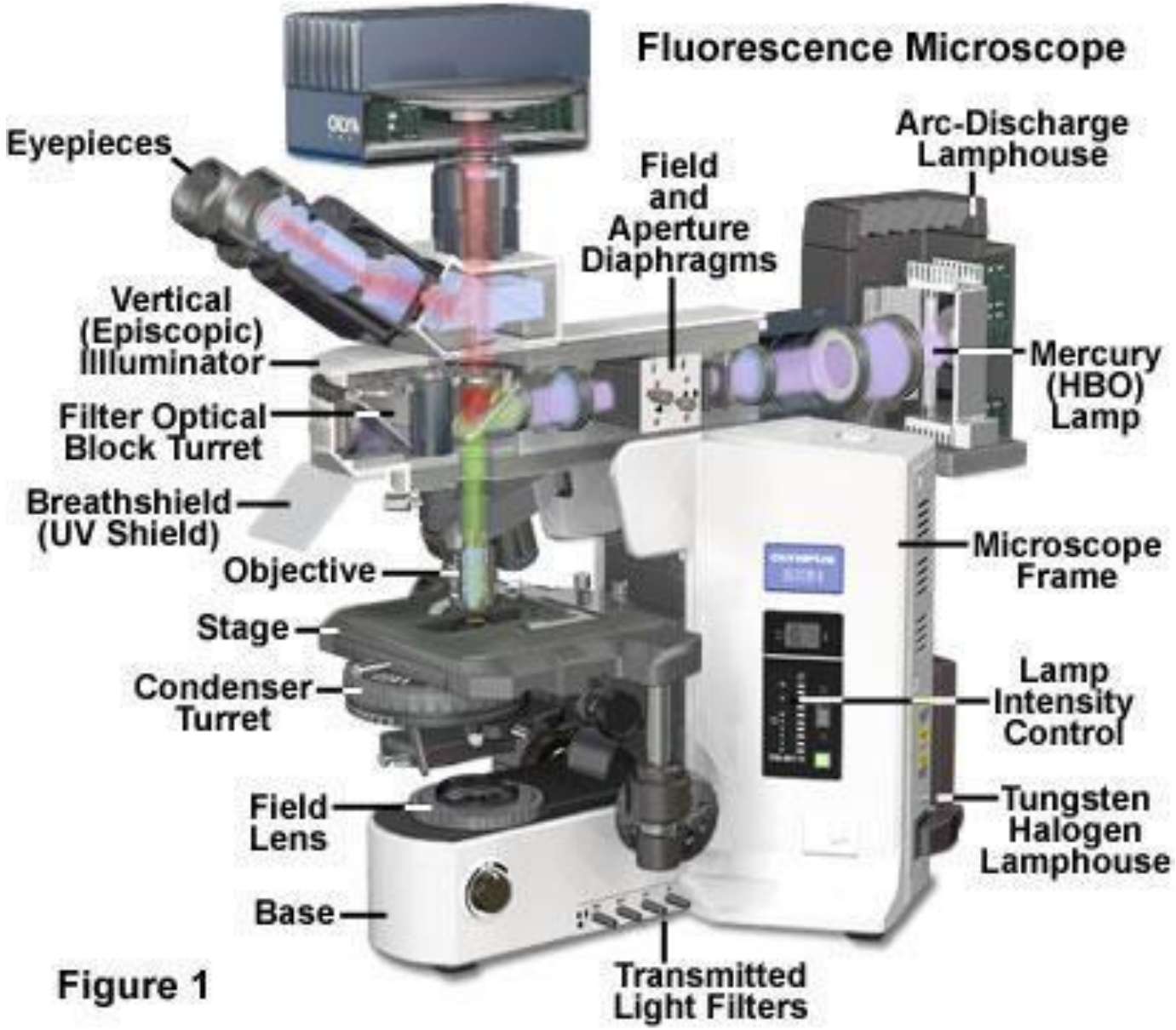


Figure 1

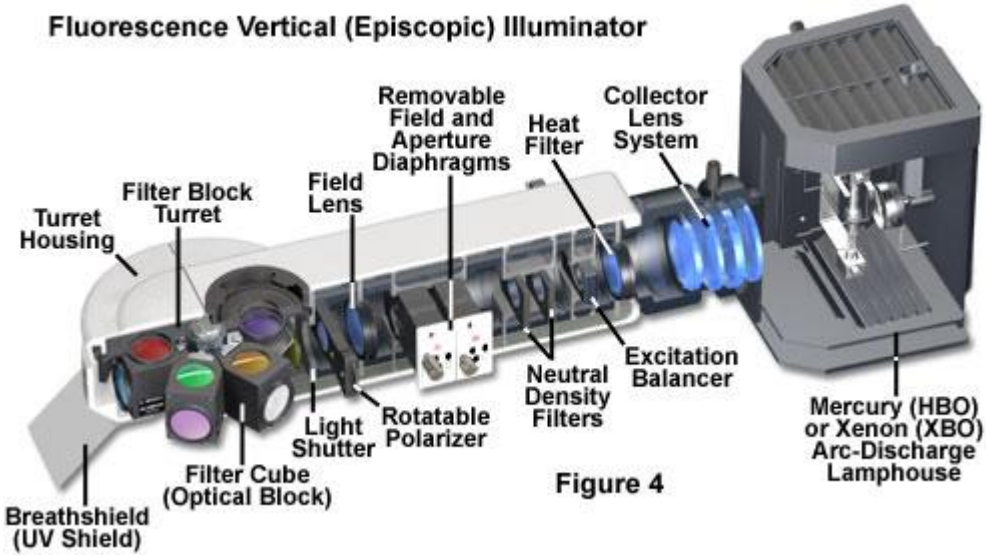


Figure 4

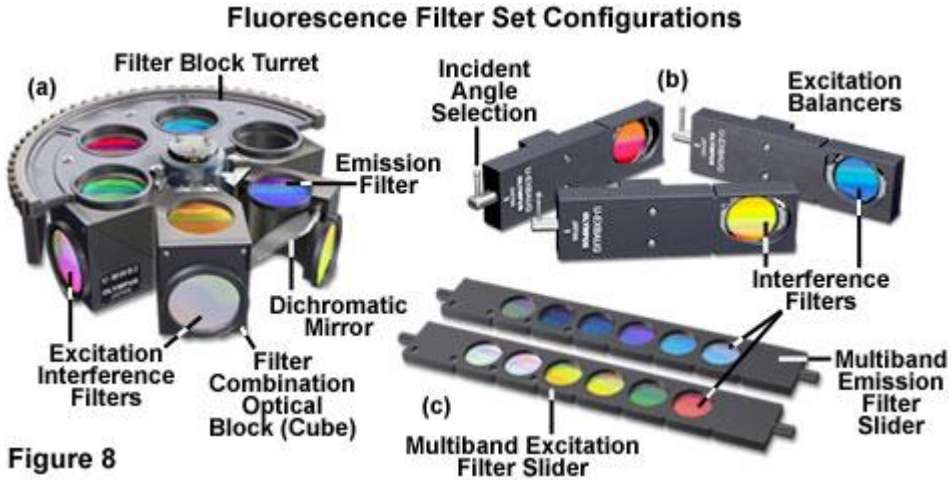


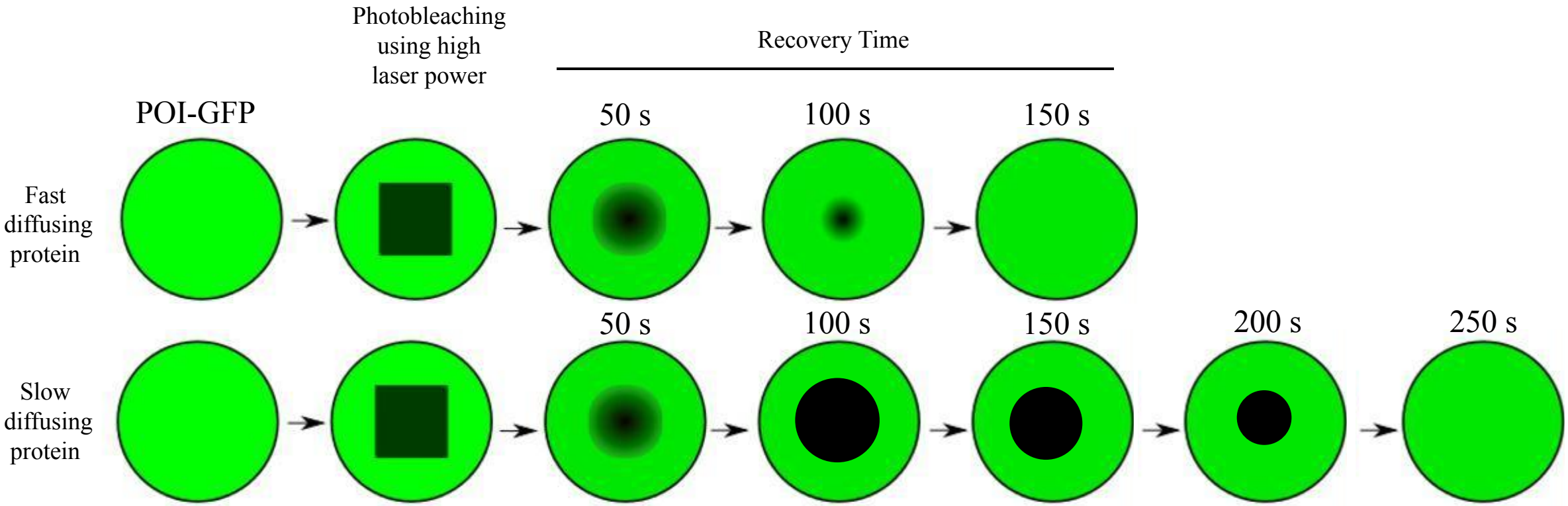
Figure 8

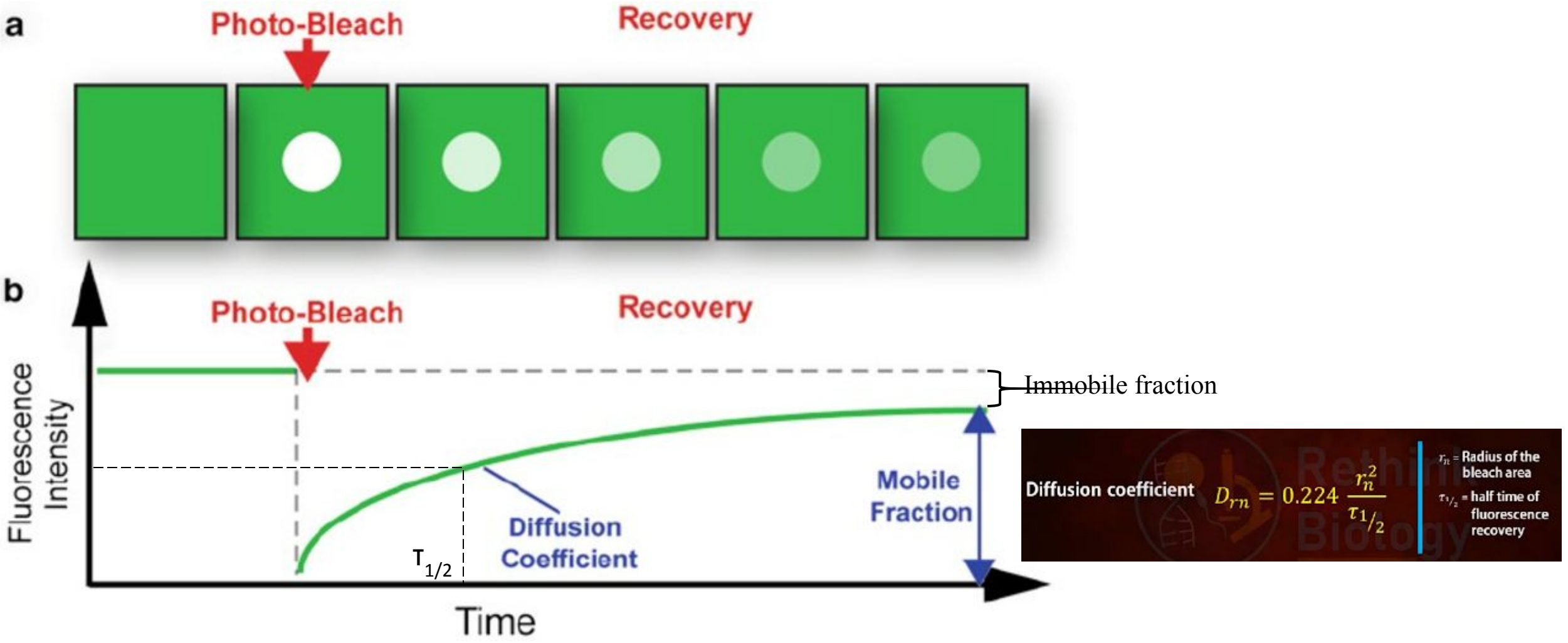
https://www.olympus-lifescience.com/en/microscope-resource/primer/techniques/fluorescence/anatomy/fluoromicroanatomy/

Fluorescence Recovery After Photobleaching (FRAP) to study protein dynamics

- FRAP was established by Axelrod et al. (1976) to study protein mobility/dynamics in living cells.
- It can be used to measure the diffusion dynamics of proteins in live cells.

POI=Protein of Interest





Mobile fraction: fraction of fluorescent protein diffusing into bleached region during the time course of the experiment.

Immobile fraction: fraction of immobile bleached molecules within the bleach spot that don't allow fluorescent molecules to occupy their space.

Aggregation of proteins, binding and diffusional barriers reduces mobile fraction.

Diffusion coefficient: Rate of movement of proteins

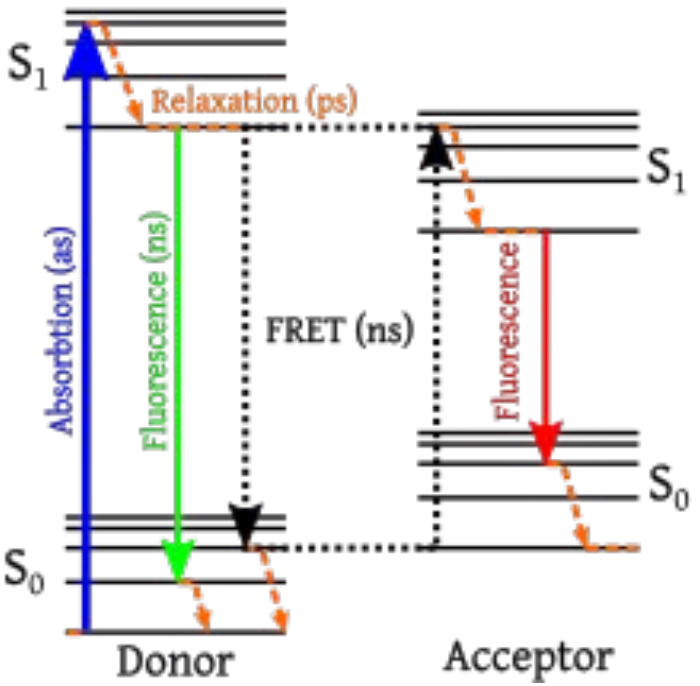
Fluorescence Resonance Energy Transfer (FRET) to study protein-protein interactions

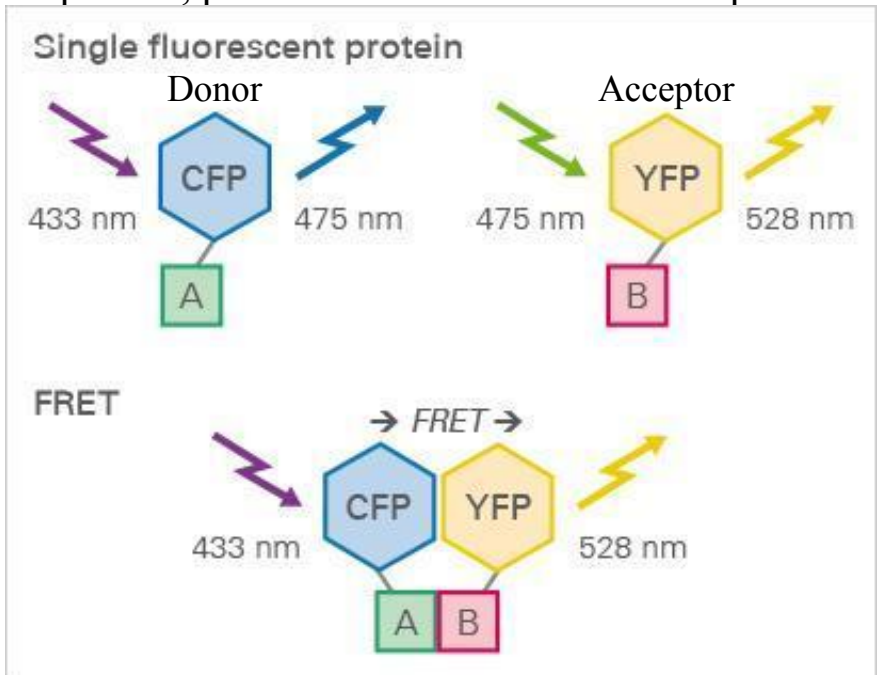
- Forster or Fluorescence Resonance Energy Transfer (FRET) is a mechanism describing energy transfer between two light-sensitive molecules (chromophores).

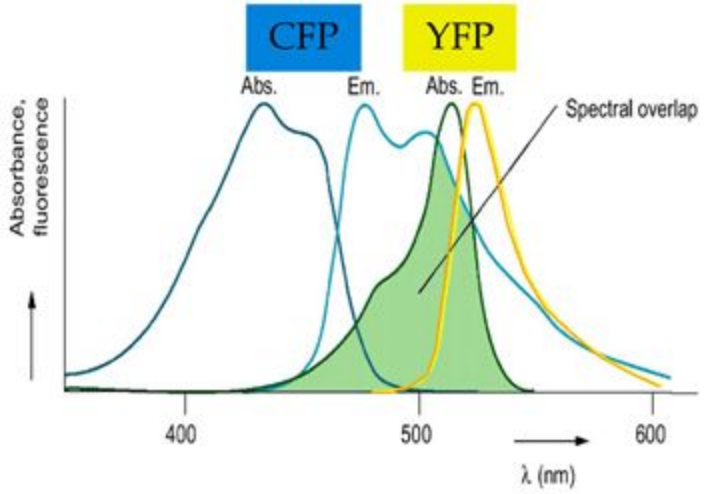
- A donor chromophore in its excited state, transfer energy to an acceptor chromophore through nonradiative dipole-dipole coupling.

- The efficiency of this energy transfer is inversely proportional to the 6th power of the distance between donor and acceptor, making FRET extremely sensitive to small change in distance (within 1-10 nm).

- FRET is used to check protein-protein, protein-DNA interactions and protein conformation dynamics.



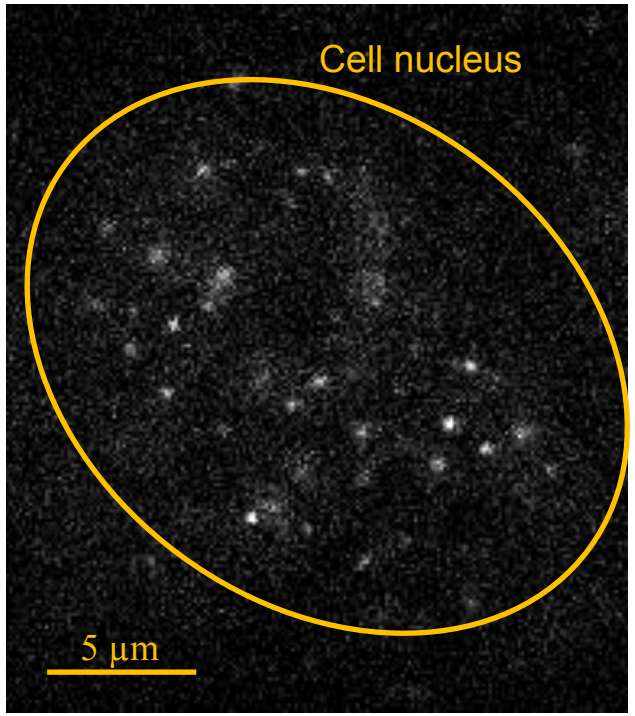




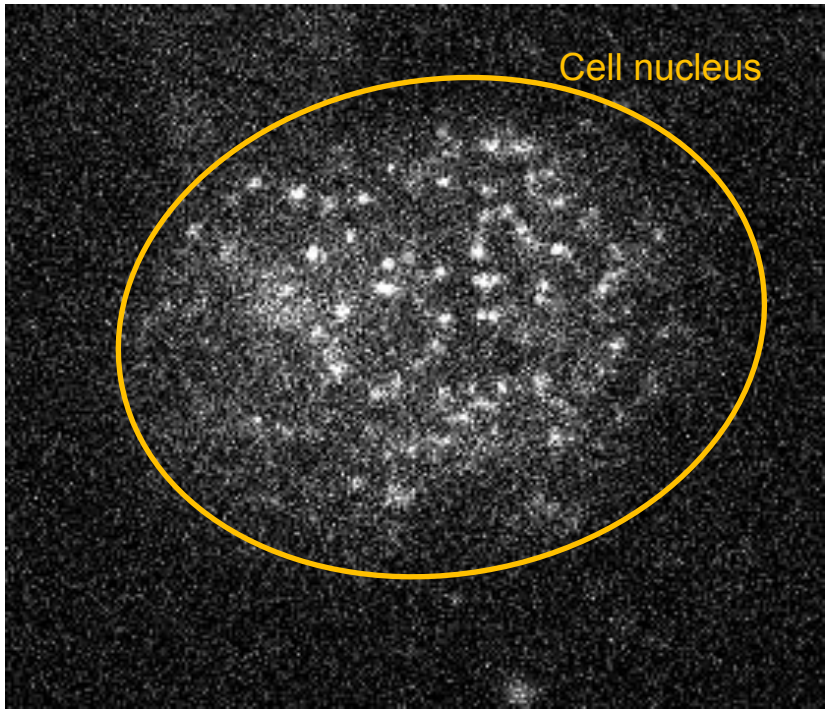
Single-Molecule Imaging to study protein dynamics

- Single-Molecule Imaging (SMI) is a method of visualizing and quantifying the dynamics of the biomolecules at the single-molecule level, either *in-vivo* or *in-vitro*.
- Classical biochemical/microscopy methods (such as IF, Western Blot, ChIP, Proteomics, FRAP etc.) describe the average behavior of biomolecules from 1000s of cells/1000s of molecules within a cell.
- Such methods don't allow the identification of subpopulations of biomolecules.
- SMI visualizes the activity of single-molecules and record it with high spatiotemporal resolution.

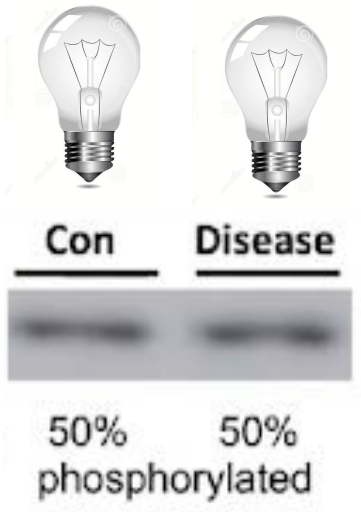
Transcription factor binding to chromatin



Histone H2B binding to chromatin

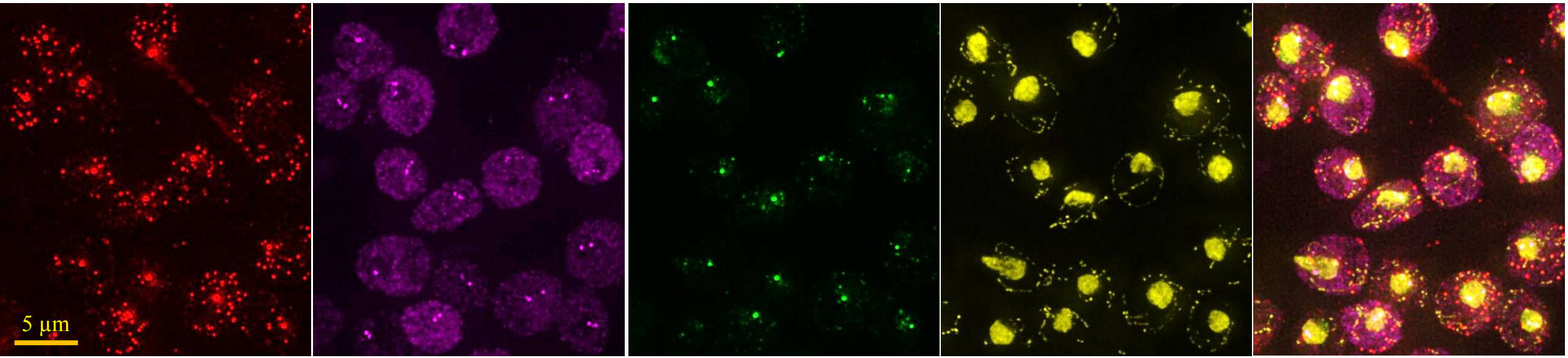


Biological processes are highly dynamic.
e.g. phosphorylation, transcription factor binding

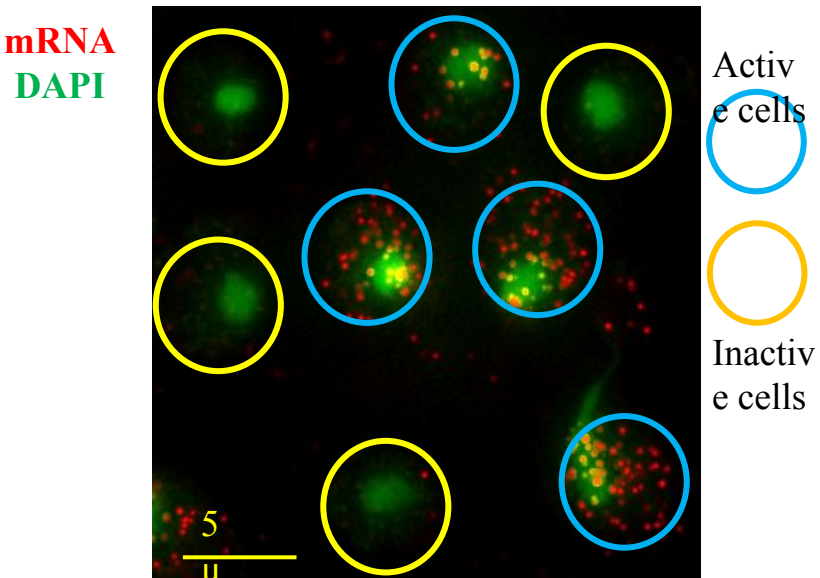


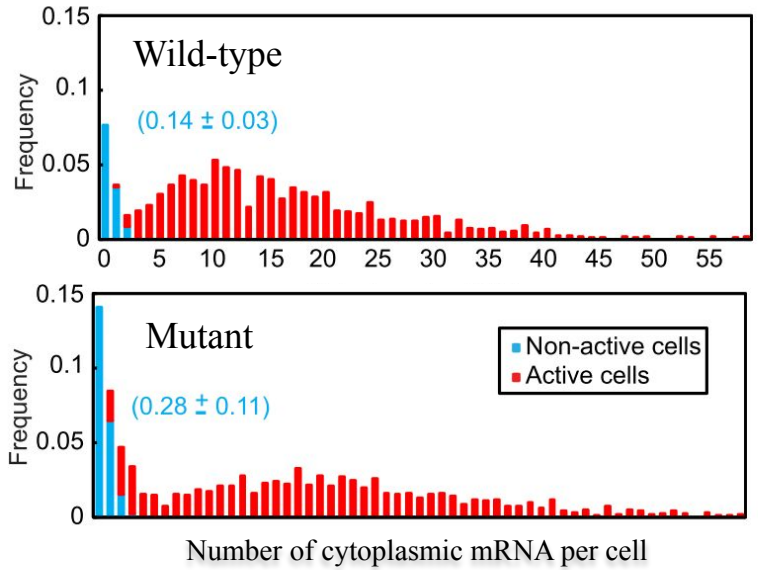
Single-Molecule FISH to quantify number of mRNA/cell

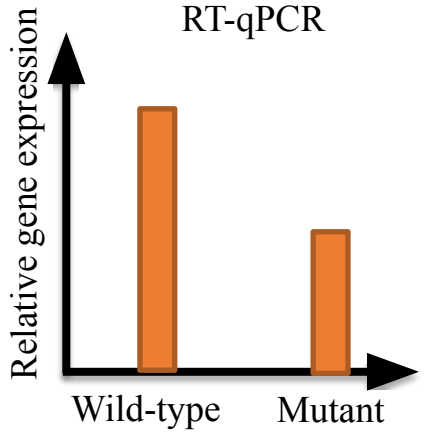
To visualize single-molecules of mRNA in fixed cells



Quantification of transcription by smFISH and RT-qPCR







Optical Tweezers to measure biological forces

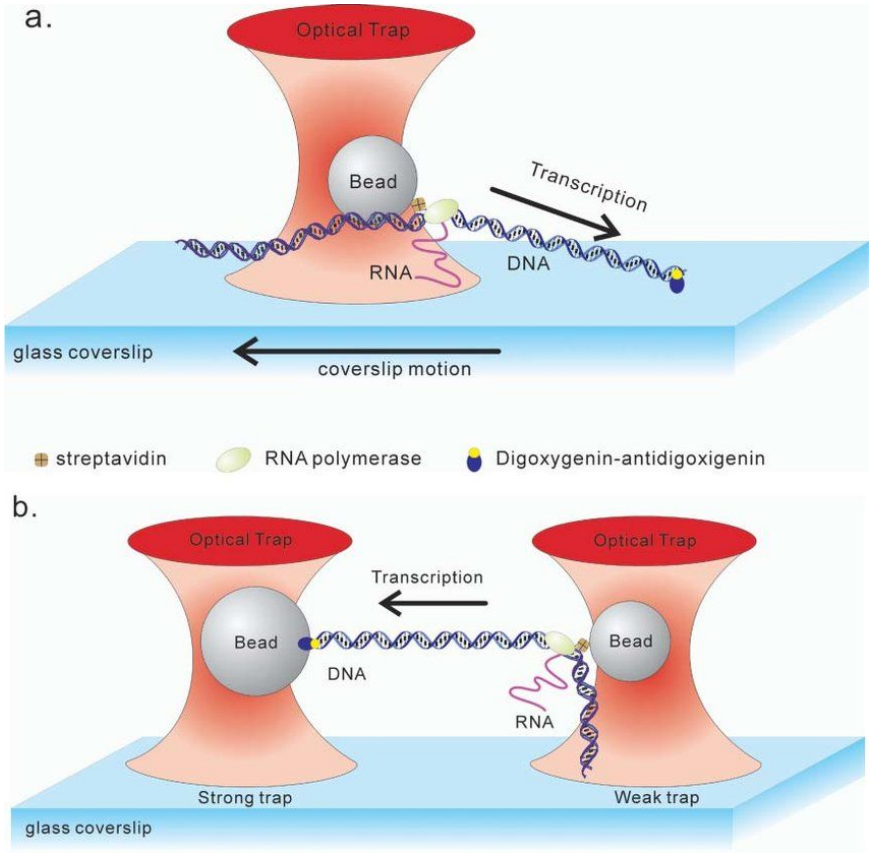
Optical tweezers use tightly focussed light beam **(IR laser) to hold and manipulate microscopically small objects.**

Microscope objective lens (high NA) is used for tight focussing of the IR laser beam. This region becomes an optical trap that can hold small objects in 3D.

Optical tweezers can also make accurate measurements of the sub-picoNewton forces exerted on the trapped objects and the small displacements (nanometer).

Since the diameter of biomolecules is generally between 1 and 10 nanometers, the optical tweezer cannot directly observe and manipulate. Hence, it is necessary to connect the molecule to the microspheres (beads) and indirectly manipulate and measure by using the beads as the "handle" of the manipulation. For example, the two ends of the DNA are connected to two beads, and the beads are manipulated by a double-beam tweezers to stretch the DNA and measure its elastic properties.

Optical tweezers have been used with great success in the field of single-molecule biophysics. For example, they have helped researchers unravel the complex elasticity and folding dynamics of DNA, RNA, proteins (especially motor protein dynamics) and other long-chain "biopolymers".

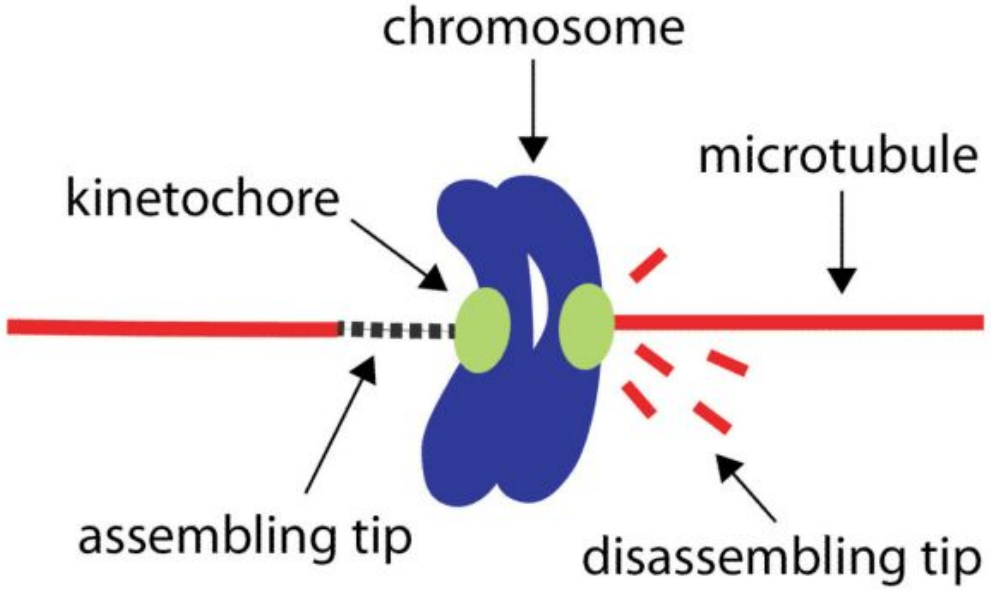


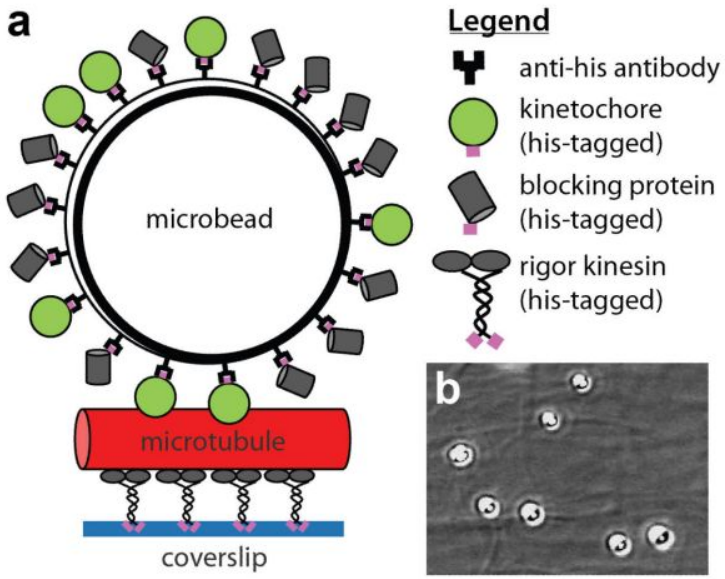
Kimura and Bianco 2006, Analyst

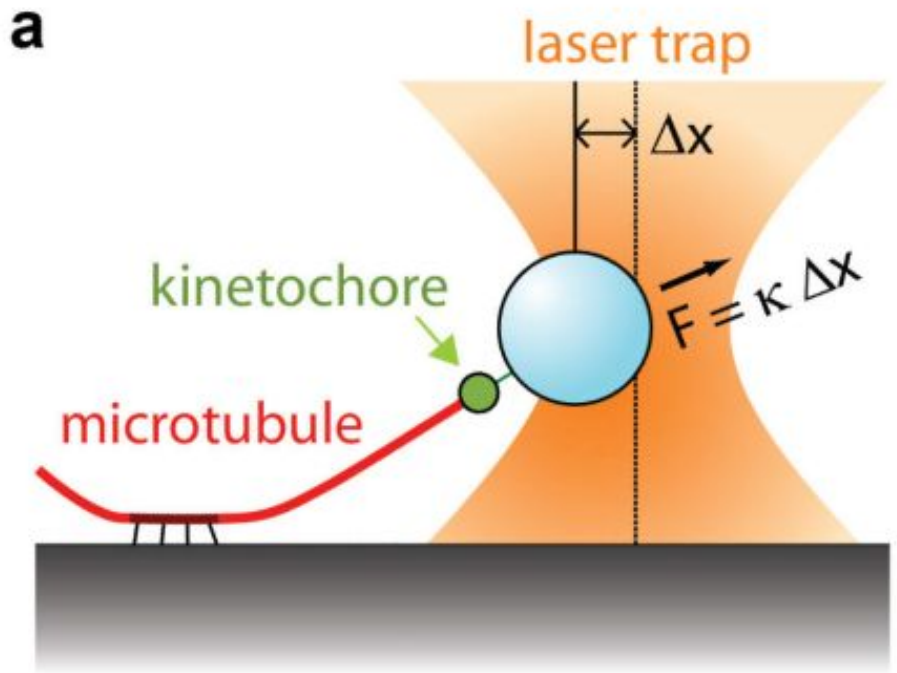
https://physicsworld.com/a/optical-tweezers-where-physics-meets-biology/

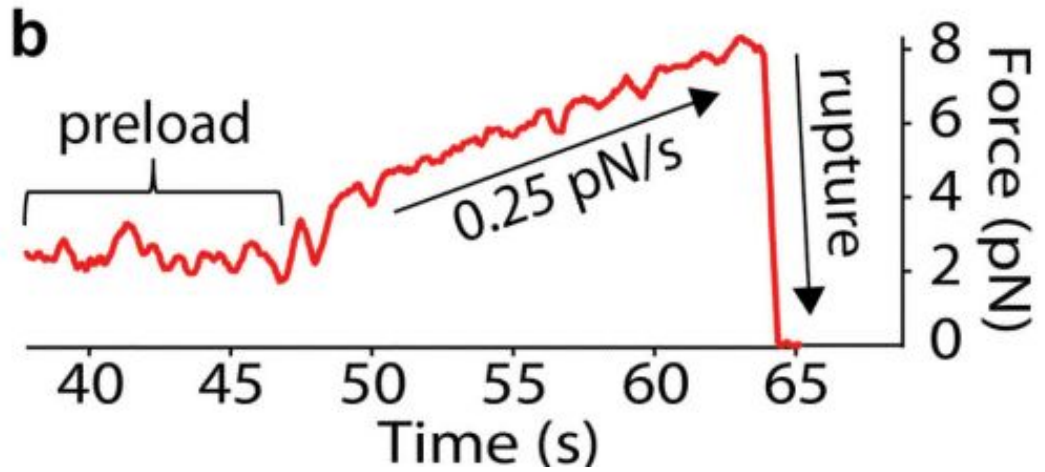
https://en.wikipedia.org/wiki/Optical_tweezers

Strength of Kinetochore-Microtubule attachments









Driver et al. 2014_Methods in Enzymology